上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

Carbon Accounting

Lecture 7 and 8

Carbon Accounting: Organizational Inventory

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饮 水 思 源 · 爱 国 荣 校



Learning Outcomes

1. **Explain** why firms conduct organizational GHG accounting
2. **Define** a simple organizational boundary and classify major emission sources into Scope 1, 2 and 3.
3. **Calculate** a simplified inventory using activity data, compatible emission factors, units and reporting periods..
4. **Interpret** BMW's inventory structure and key assumptions.

Lecture 7

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Internal vs External

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GHGP Corporate Standard, ISO 14064-1

Lecture 8

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TBA

02- Wrap up and Next time

* In-class Project activity

Appendix A

Greenhouse gas emissions

Apple Inc.

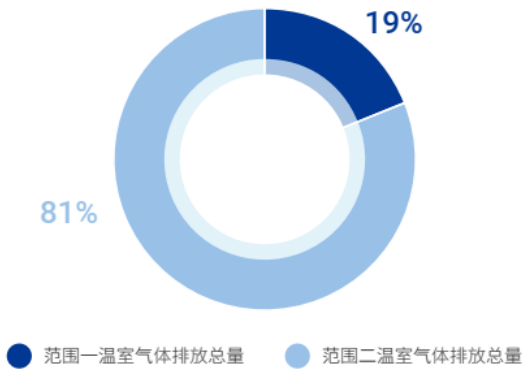
We account for our carbon footprint by following internationally recognized standards, like the World Resources Institute (WRI) Greenhouse Gas (GHG) Protocol and ISO 14040/14044.[1] Improving the accuracy of our carbon footprint is an ongoing process — as we learn more, we refine our carbon models and adjust our climate roadmap. We also regularly revisit the boundary of our carbon footprint as our data sources improve and our business evolves. For example, in 2022, we expanded our corporate footprint to include work from home emissions, third-party cloud services, electricity transmission and distribution losses, and upstream impacts from our scope 1 fuel use.

		Fiscal year				
		2022	**2021**	**2020**	**2019**	**2018**
Corporate emissions (metric tons CO_2e)[2]	Gross emissions	**324,100**	**166,380**	**334,430**	**573,730**	**586,170**
	Scope 1	**55,200**	**55,200**	**47,430**	**52,730**	**57,440**
	Natural gas, diesel, propane	39,700	40,070	39,340	40,910	42,840
	Fleet vehicles	12,600	12,090	4,270	6,950	11,110
	Other emissions[3]	2,900	3,040	3,830	4,870	3,490
	Scope 2 (market-based)[4]	**3,000**	**2,780**	**0**	**0**	**8,730**
	Electricity	0	0	0	0	8,730
	Steam, heating, and cooling[5]	3,000	2,780	0	0	0
	Scope 3	**265,800**	**108,400**	**287,000**	**521,000**	**520,000**
	Business travel	113,500	22,850	153,000	326,000	337,000
	Employee commute[6]	134,200	85,570	134,000	195,000	183,000
	Upstream fuel	10,600	0	0	0	0
	Work from home (market-based)	7,500	0	0	0	0
	Transmission and distribution loss (market-based)	0	N/A	N/A	N/A	N/A
	Third-party cloud (market-based)	0	0	0	0	0
	Carbon removals					
	Corporate carbon offsets[7]	–324,100	–167,000[8]	–70,000[9]	0	0
Product life cycle emissions (metric tons CO_2e)[10]	Gross emissions (Scope 3)	**20,280,000**	**23,020,000**	**22,260,000**	**24,460,000**	**24,550,000**
	Manufacturing (purchased goods and services)	13,400,000	16,200,000	16,100,000	18,900,000	18,500,000
	Product transportation (upstream and downstream)	1,900,000	1,750,000	1,800,000	1,400,000	1,300,000
	Product use (use of sold products)	4,900,000	4,990,000	4,300,000	4,100,000	4,700,000
	End-of-life processing	80,000	80,000	60,000	60,000	50,000
	Carbon removals					
	Product carbon offsets	0	–500,000[11]	0	0	0
Total gross scope 3 emissions (corporate and product) (metric tons CO_2e)		**20,550,000**	**23,128,400**	**22,547,000**	**24,980,000**	**25,070,000**
Total gross carbon footprint (without offsets) (metric tons CO_2e)[12]		**20,600,000**	**23,200,000**	**22,600,000**	**25,100,000**	**25,200,000**
Total net carbon footprint (after applying offsets) (metric tons CO_2e)[12]		**20,300,000**	**22,530,000**	**22,530,000**	**25,100,000**	**25,200,000**

表 4 组织边界内公司及子公司 2022 年度温室气体排放量

温室气体排放类别	单位	2022年度排放量
范围一温室气体排放	吨二氧化碳当量	610,885.46
范围二温室气体排放	吨二氧化碳当量	2,631,947.26
范围一 + 范围二温室气体排放	吨二氧化碳当量	3,242,832.72

按范围划分的排放占比



按产品类型[3]划分的排放占比

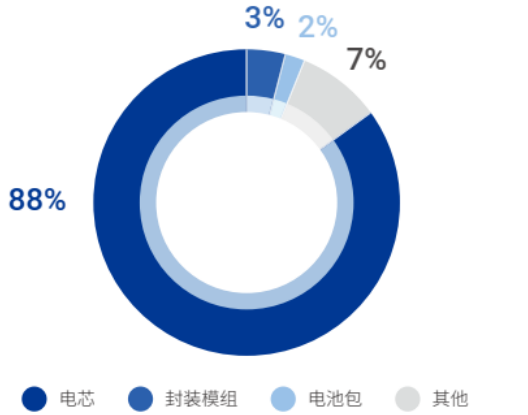
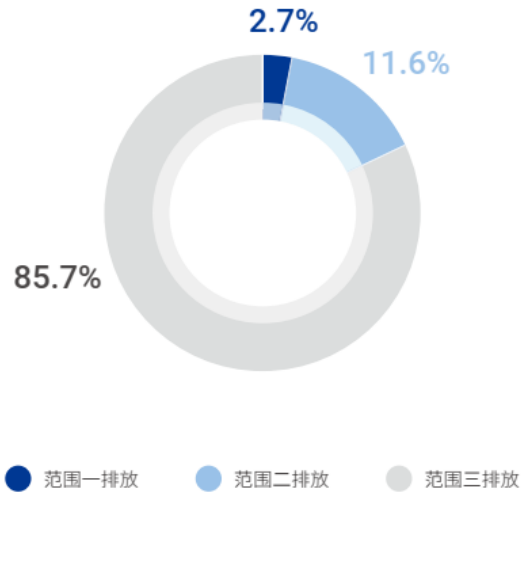


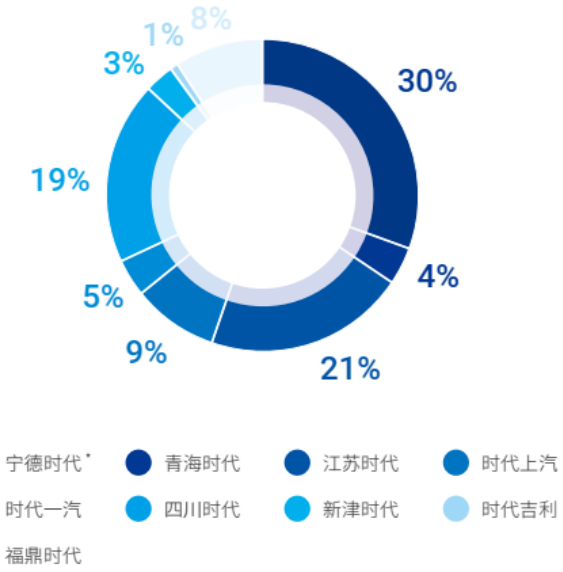
表 5 已核查的 9 家生产基地基地 2022 年度温室气体排放量

温室气体排放类别	单位	2022年度排放量
范围一温室气体排放	吨二氧化碳当量	451,783.78
范围二温室气体排放	吨二氧化碳当量	1,981,516.06
范围三温室气体排放	吨二氧化碳当量	14,632,401.23
范围一 + 范围二温室气体排放	吨二氧化碳当量	2,433,299.84
间接（范围二 + 范围三）温室气体排放	吨二氧化碳当量	16,613,917.29
范围一 + 范围二 + 范围三温室气体总排放	吨二氧化碳当量	17,065,701.06

按范围划分的排放占比



按生产基地划分的排放[5]占比



来源：https://www.catl.com/about/sustainability/

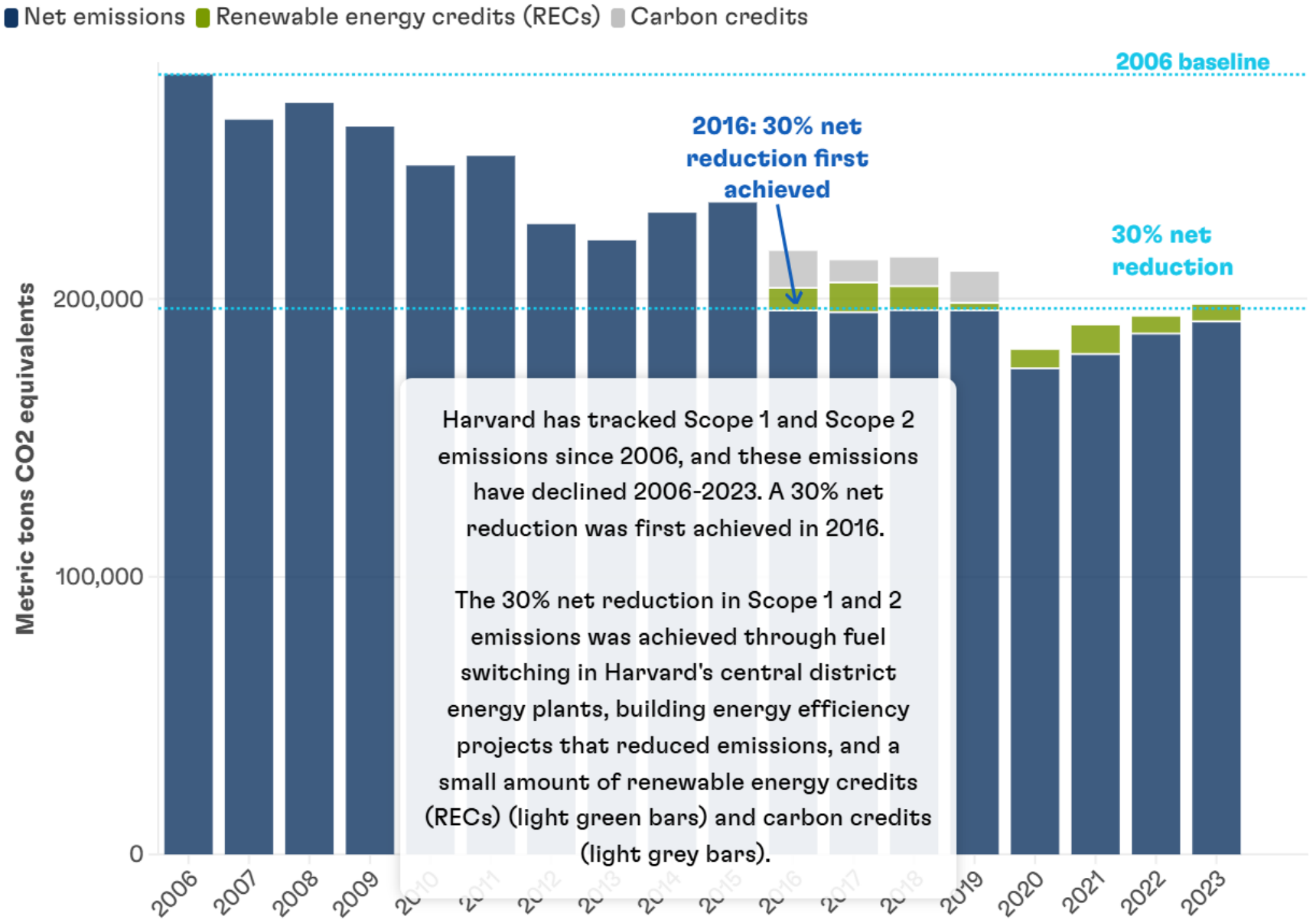
表7-1 国际校区温室气体排放量

GHG Emission of the International Campus (Table 7-1)

运营边界 Operational boundaries	主要排放源 Main Emission Source	能源类别 Types of Energy	计量单位 Unit	报告年消耗量 Con-sumption	CO_2	CH_4	N_2O
范围1: 直接温室气体排放 Scope 1: direct GHG emissions	锅炉 Boiler	天然气 Natural Gas	10k NM^3	14.9	311.41	0.0056	0.0006
	食堂餐厅 Dining Hall	天然气 Natural Gas	10k NM^3	7	146.30	0.0026	0.0003
	校车、公务车 Shuttle bus, business vehicles	柴油 Diesel	10k L	2.29	62.57	0.0033	0.0033
	校车、公务车 Shuttle bus, business vehicles	汽油 Petrol	10k L	2.92	65.95	0.0238	0.0076
范围2: 间接温室气体排放 Scope 2: indirect GHG emissions	照明、空调、动力设施等 Light, A/C and other facilities	外购电 Electricity	10k kWh	2199.84	17845.09	0.0000	0.0000

From: “2022 Accounting Report on Carbon Emission for International Campus, Zhejiang University”

Net Emissions Trend



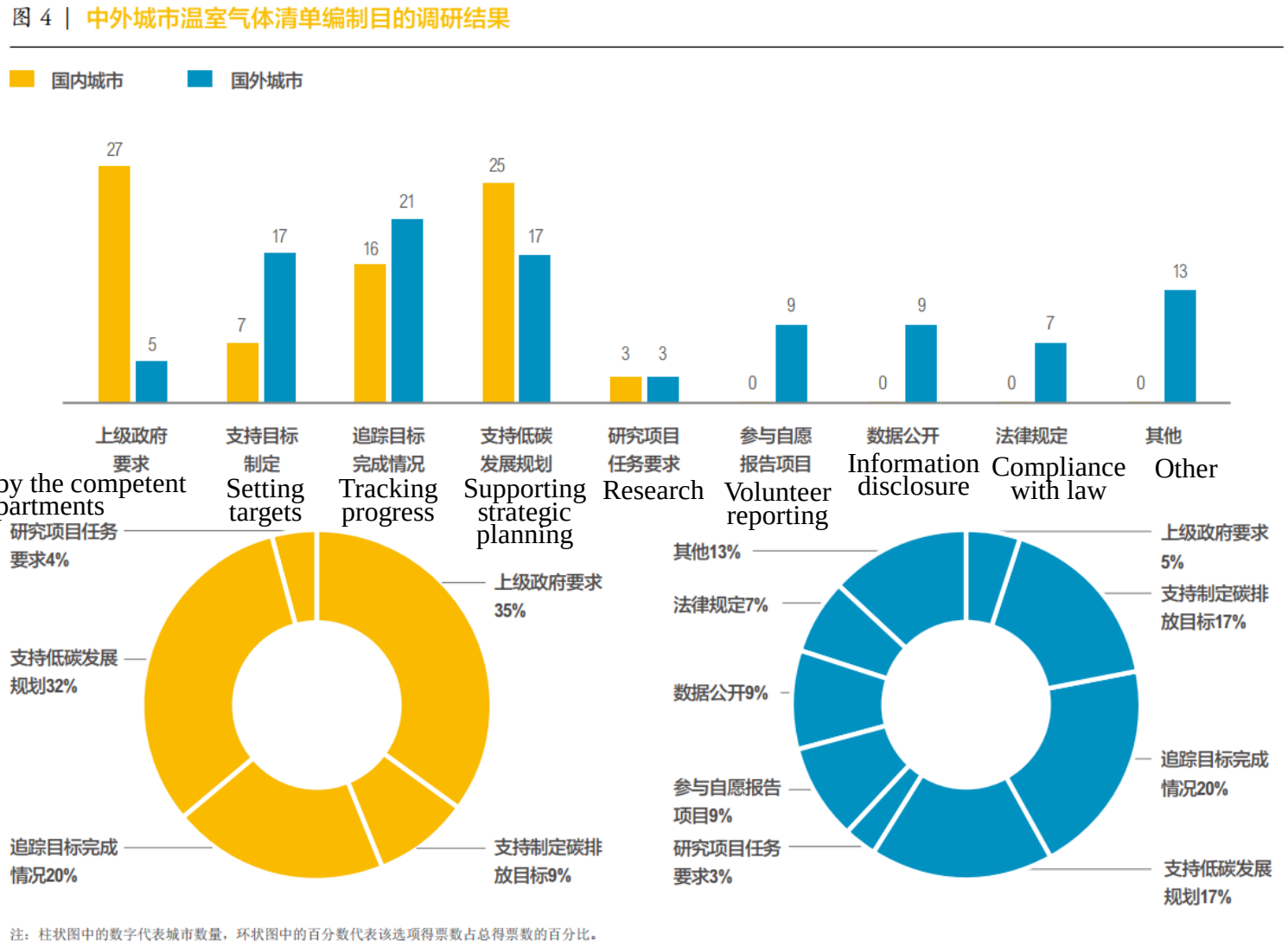
HARVARD
Office for Sustainability

01

Why Companies Account for GHG Emissions?

Internal vs External motivations

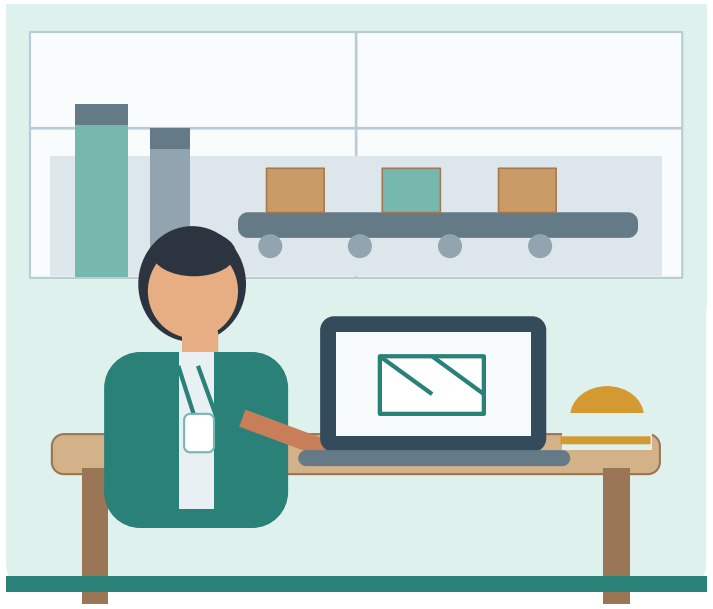
Review: Purposes of city-level inventory



Source: 中国城市温室气体清单应用的领域、挑战和建议. 北京：世界资源研究所.
https://wri.org.cn/research/uses-challenges-and-recommendations-application-city-greenhouse-gas-inventories-china

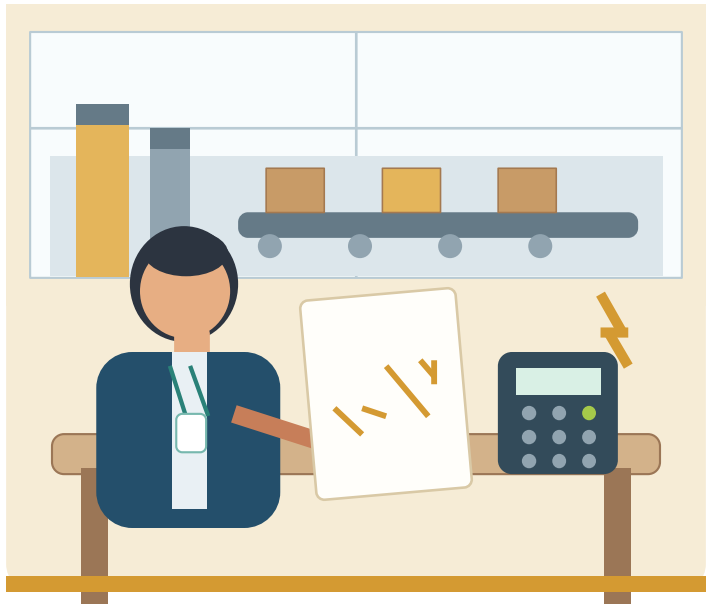
Organizational-level accounting – what it is for?





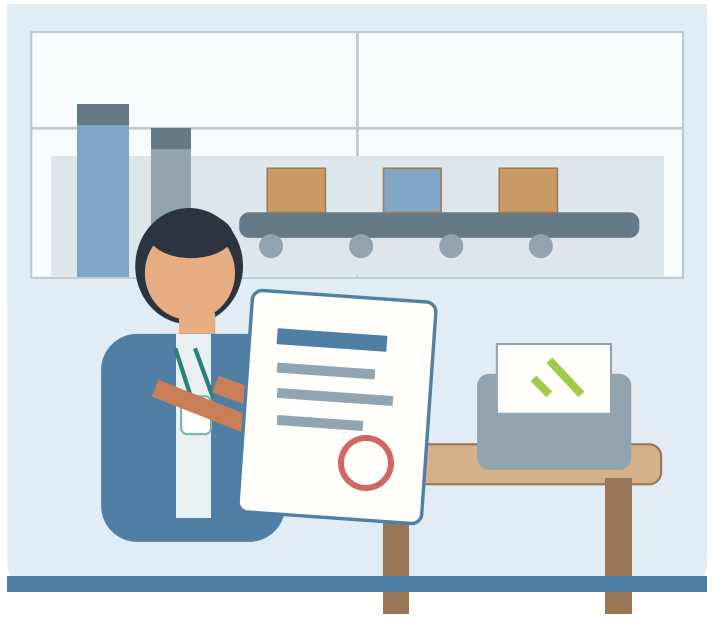
Customer

What are your company's annual GHG emissions?



Finance team

Where can we cut emissions and energy costs?



Regulator

Can you submit the required emissions report?

Pictures created by ChatGPT-5.6 Sol

Why Companies Account for GHG Emission?

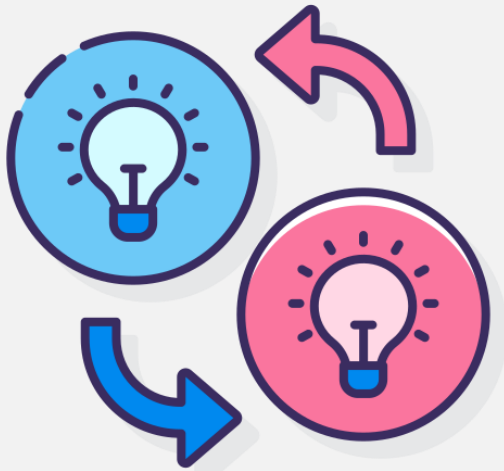
Think

1 min



Pair

2 min



Share

1 min

Yuketang Voting



Why do companies make effort to quantify GHG emissions?

投票 最多可选3项

设置

In your opinion, what are the **three** primary motivations for enterprises to carry out organizational carbon accounting?

A Regulatory or policy requirements

B Customer and supply-chain requirements

C Internal decarbonization and cost reduction

D Brand image and corporate responsibility

E Carbon asset development and carbon credit revenue

F Financing and ESG rating needs

G Identify climate-related risk and support Long-term sustainable strategy

提交



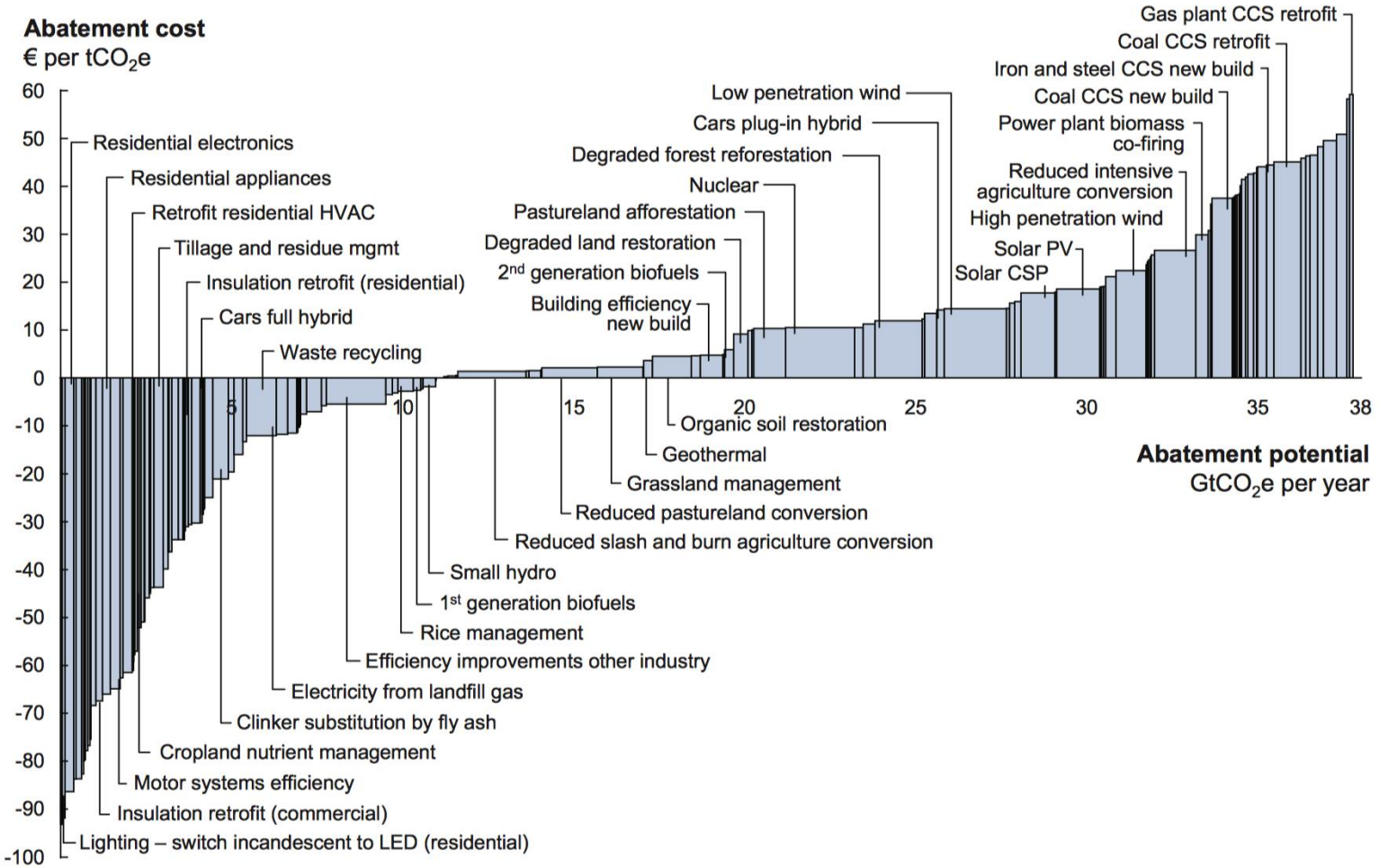
"Measuring and disclosing GHG emissions can be a key tool in achieving sustainability goals, but the strategic value of climate-related reporting depends on measuring what matters and using what is learned to advance a vision for change. "

Counting What Counts: How Carbon Accounting Can Drive Climate Strategy, Value - WSJ

Cost savings

Exhibit 1

Global GHG abatement cost curve beyond business-as-usual – 2030

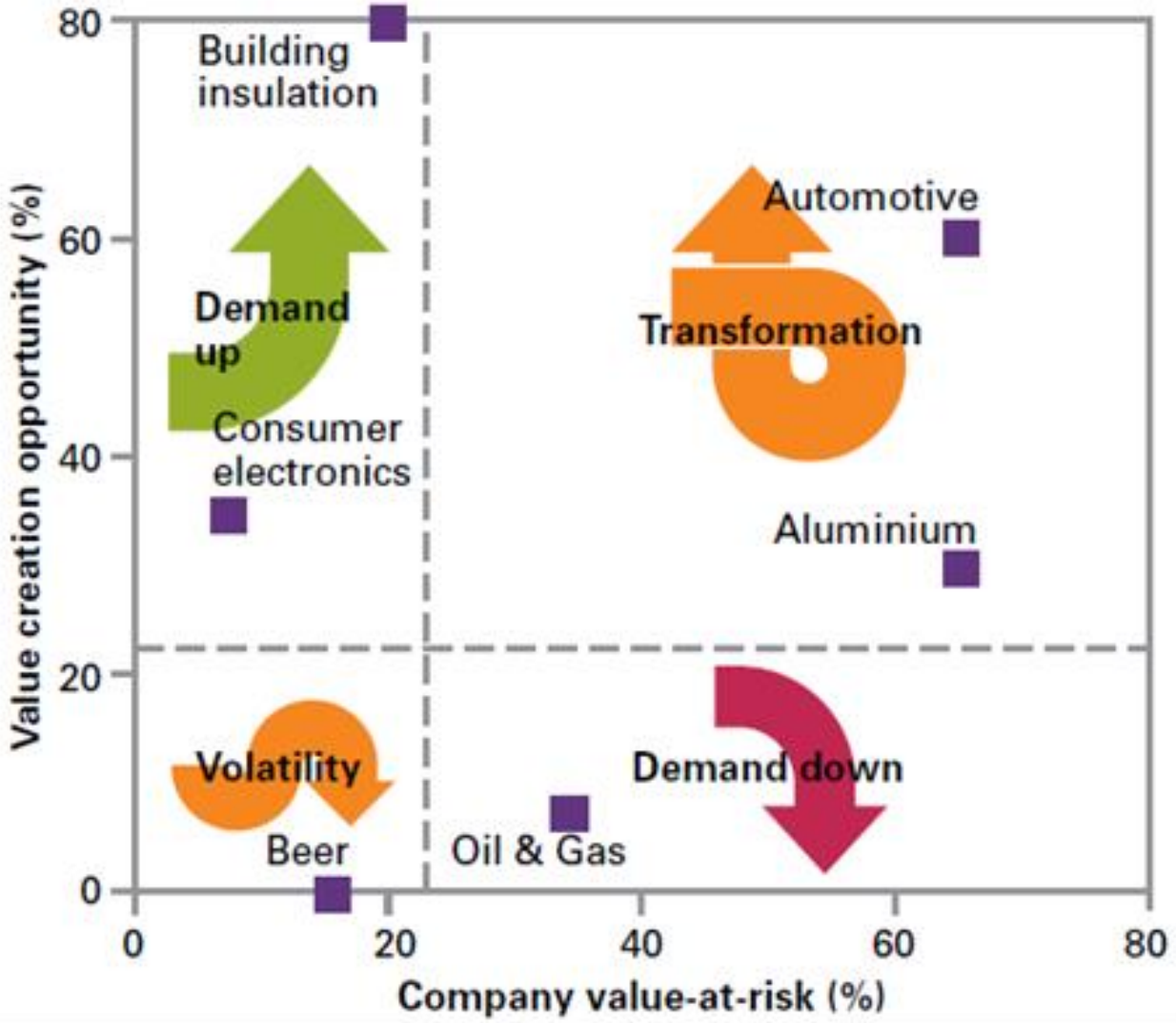


Note: The curve presents an estimate of the maximum potential of all technical GHG abatement measures below €60 per tCO_2e if each lever was pursued aggressively. It is not a forecast of what role different abatement measures and technologies will play.
Source: Global GHG Abatement Cost Curve v2.0

(Source: Mckinsey & Company

https://www.mckinsey.com/business-functions/sustainability/our-insights/greenhouse-gas-abatement-cost-curves)

Understanding opportunities and risks



"…companies must be able to understand and manage their GHG risks if they are to ensure long-term success in a competitive business environment" (GHG Protocol, Corporate Standard).



Carbon Exposure

Vs

Risk Exposure

(Source: Carbon Trust - https://www.carbontrust.com/media/84956/ctc740-climate-change-a-business-revolution.pdf)

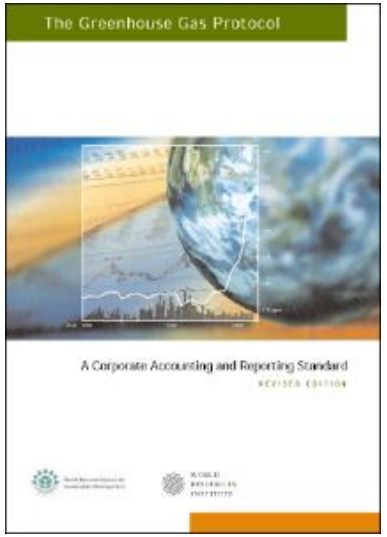
02

Main Standards and How It is Done

Introduction to International Standards

- The GHG Protocol: A Corporate Accounting and Reporting Standard

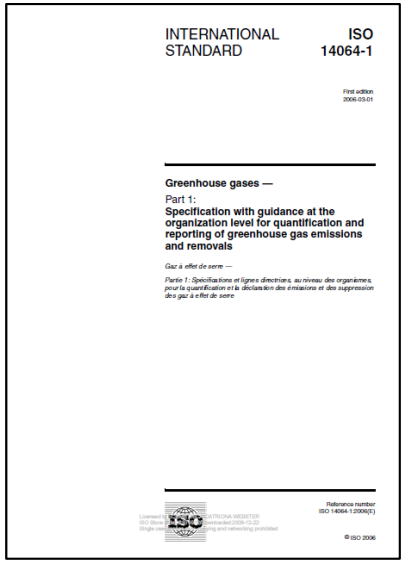
(Source: http://www.ghgprotocol.org/files/ghgp/public/ghg-protocol-revised.pdf)

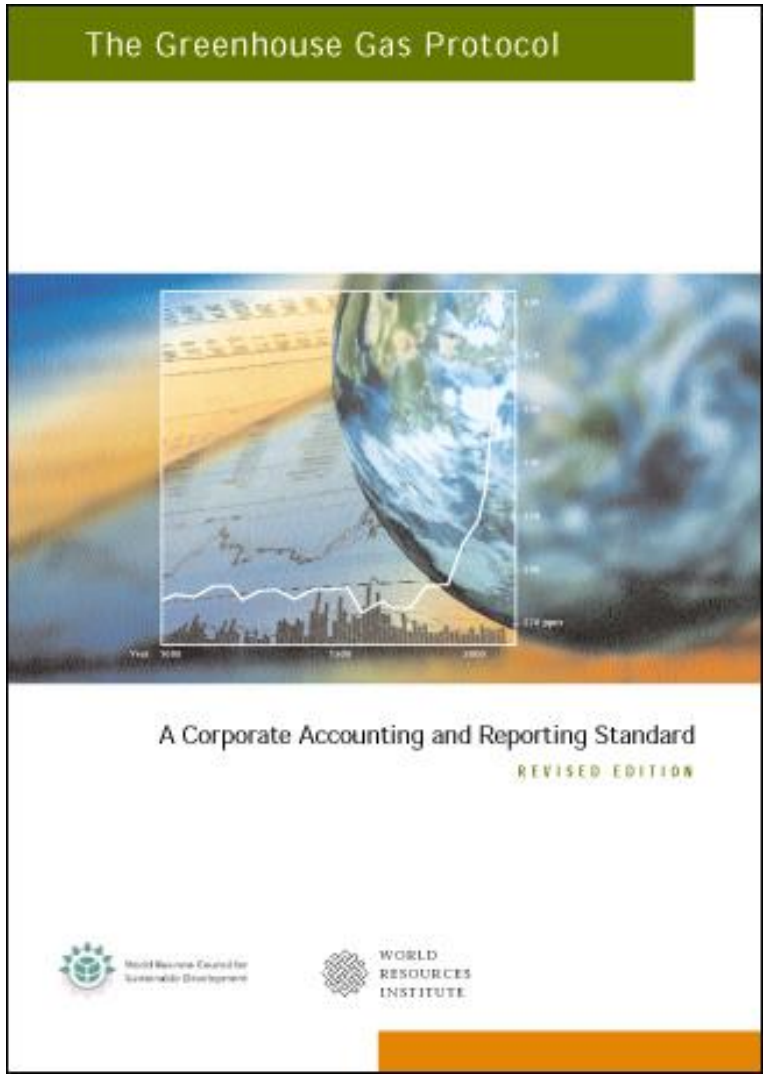


(2025.9)

- ISO 14064 – Part 1 Specification with guidance at the organization level for quantification and reporting of greenhouse gas emissions and removals

(Source: http://www.iso.org/iso/catalogue_detail?csnumber=38381)





- The GHG Protocol:
  - Partnership of businesses, NGOs, and others
  - Convened by World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD)

  http://www.ghgprotocol.org/

- The GHG Protocol: A Corporate Accounting and Reporting Standard:
  - First published in 2001
  - Revised in 2004, updated in 2015
  - A third version is expected in 2027

Most popular standard – 92% of Fortune 500 companies reporting to the CDP used the GHG Protocol in 2016

GHG Protocol is currently revising its corporate suites of standards and guidance. The updates can be seen at https://ghgprotocol.org/ghg-protocol-corporate-suite-standards-and-guidance-update-process

GHG Protocol's Principles

- These principles underpin all aspects of GHG accounting and reporting and are derived in part from generally accepted financial accounting principles
  1. **Relevance:** ensure the GHG inventory appropriately reflects the emissions of the company and serves the decision-making needs of the users (internally & externally)
  2. **Completeness:** are all relevant emission sources within the inventory boundary accounted for in a meaningful and comprehensive manner
  3. **Consistency:** use of consistent methodologies to allow for meaningful comparisons of emissions over time
  4. **Transparency:** address all relevant issues in a factual and coherent manner based on a clear audit trail
  5. **Accuracy:** data should be sufficiently precise to enable intended users to make decisions with reasonable assurance

Main steps in organizational GHG accounting

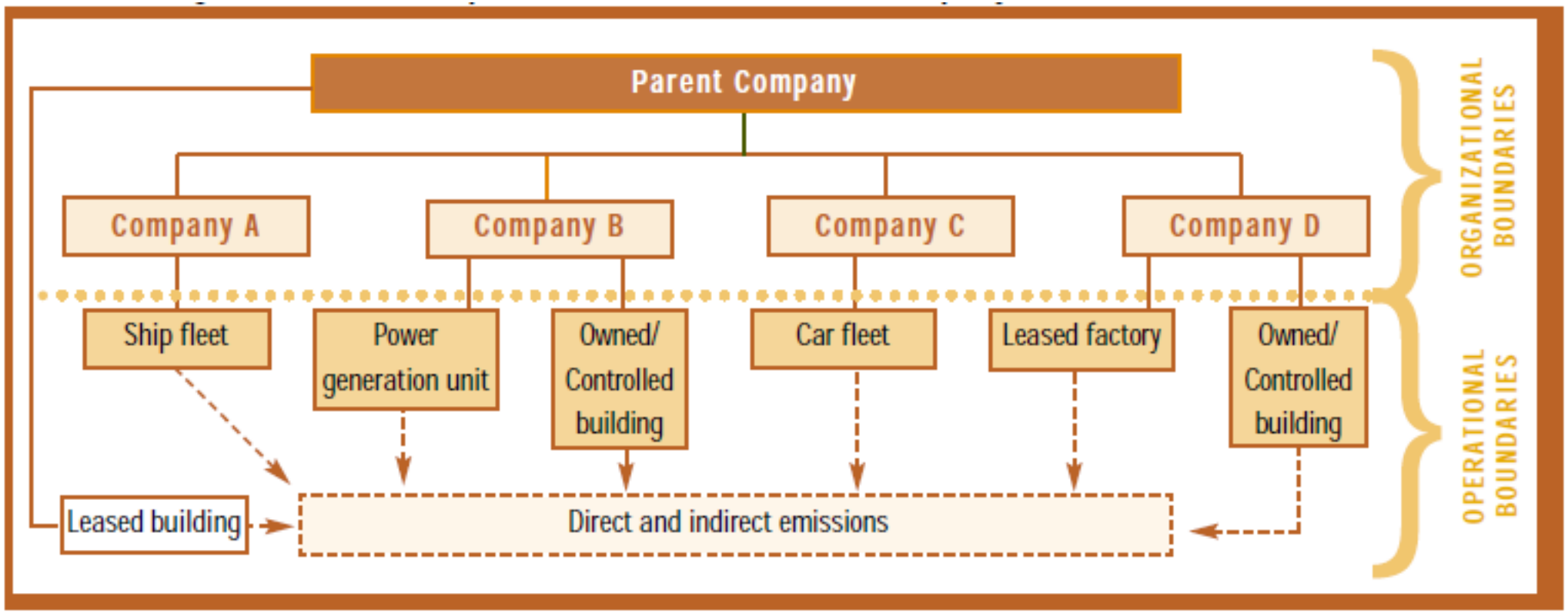
1. Set organizational boundary
2. Set operational boundary
3. Define time period, e.g. calendar year
4. Collect activity data and emission factors
5. Apply emission factors and convert to CO_2e

6. Reporting

- **Other steps**

  Target setting, Base year recalculation, Assurance and verification

Organizational and operational boundaries



Step 1: set organizational boundaries

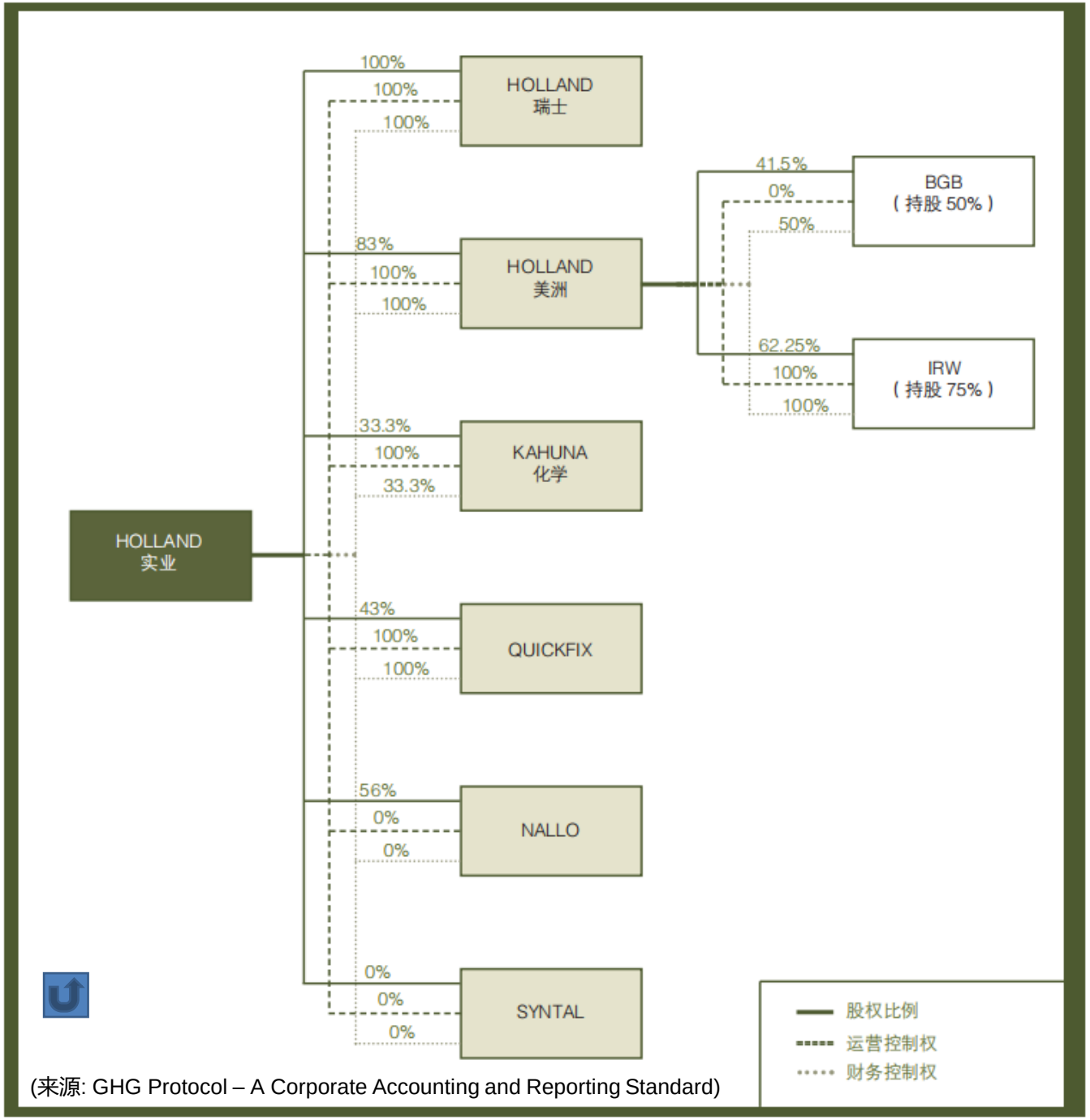
1. **Operational Control** (59%*): has full authority to introduce and implement its operating policies at the operation
    - 100% of emissions from operations which a company has operational control

2. **Financial Control** (29%*): possesses the ability to direct financial and operating policies of the latter with a view to gain economic benefits from its activities
    - 100% of emissions from operations which a company has financial control
    - One exception exists in the case of joint ventures/partnerships/operations where partners have joint financial control. (In this case equity share of GHG emissions are used.) (GHG Protocol pp. 19)

3. **Equity Share** (4%*): reflects the economic interest (extent of rights a firm has to the risks and rewards of an operation)
    - Typically utilized only by the Oil and Gas and Utilities industries.

* Percentages based on 1595 public respondents to the 2011 CDP (6% were marked “Other” & 2% were blank)

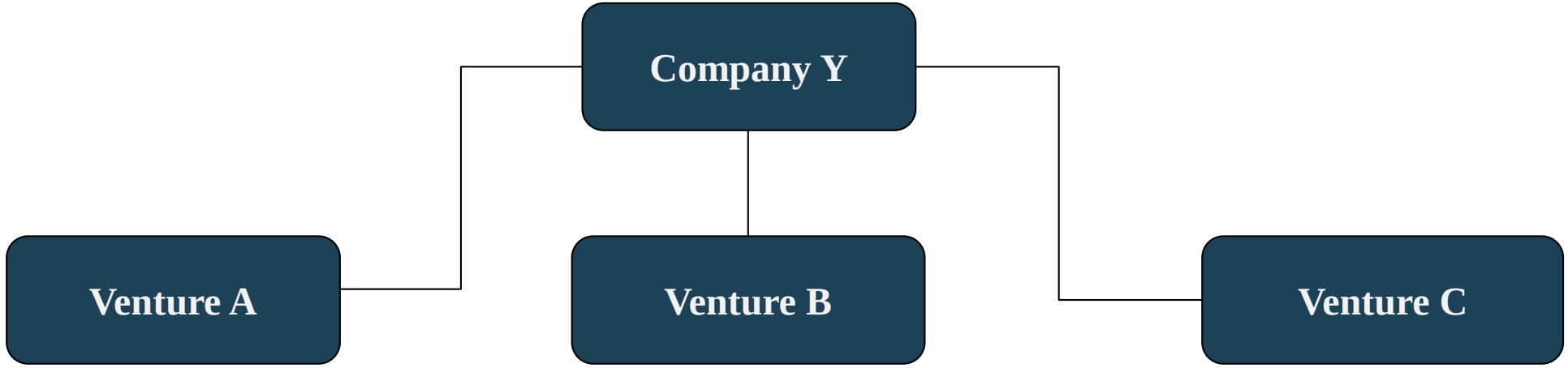
图 1　设定 Holland 实业的组织边界



(来源: GHG Protocol – A Corporate Accounting and Reporting Standard)

表 2　Holland 实业——组织结构与温室气体排放量核算

Holland 的全资与合营方	法律结构与合作方	Holland 实业持有的股权	运营政策控制权	Holland 实业财务核算处理（见表 1）	Holland 实业排放量的核算与报告	
					股权比例法	控制权法
Holland 瑞士	公司法人	100%	Holland 实业	全资子公司	100%	运营控制权 100% 财务控制权 100%
Holland 美洲	公司法人	83%	Holland 实业	子公司	83%	运营控制权 100% 财务控制权 100%
BGB	合资企业，合作方共同控制财务，另一方为 Rearden	Holland 美洲持有 50%	Rearden	通过 Holland 美洲处理	41.5% （83% × 50%）	运营控制权 0% 财务控制权 50% （50% × 100%）
IRW	Holland 美洲的子公司	Holland 美洲持有 75%	Holland 美洲	通过 Holland 美洲处理	62.25% （83% × 75%）	运营控制权 100% 财务控制权 100%
Kahuna 化学	非法人合资企业；合作方共同控制财务；另两个合作方：ICT 和 BCSF	33.3%	Holland 实业	按比例合并的合资企业	33.3%	运营控制权 100% 财务控制权 33.3%
QuickFix	合资法人，另一合作方为 Majox	43%	Holland 实业	子公司（Holland 实业享有财务控制权，因为在财务账目上其将 QuickFix 作为子公司处理）	43%	运营控制权 100% 财务控制权 100%
Nallo	合资法人，另一合作方为 Nagua 公司	56%	Nallo	关联公司（Holland 实业没有财务控制权，因为在财务账目上其将 Nallo 作为关联公司处理）	56%	运营控制权 0% 财务控制权 0%
Syntal	公司法人，Erewhon 公司的子公司	1%	Erewhon 公司	固定资产投资	0%	运营控制权 0% 财务控制权 0%



- 80% equity share
- Operational control
- No financial control

- 30% equity share
- No operational control
- Financial control

- 100% equity share
- Operational control
- Financial control

Operational Control: Venture A and Venture C

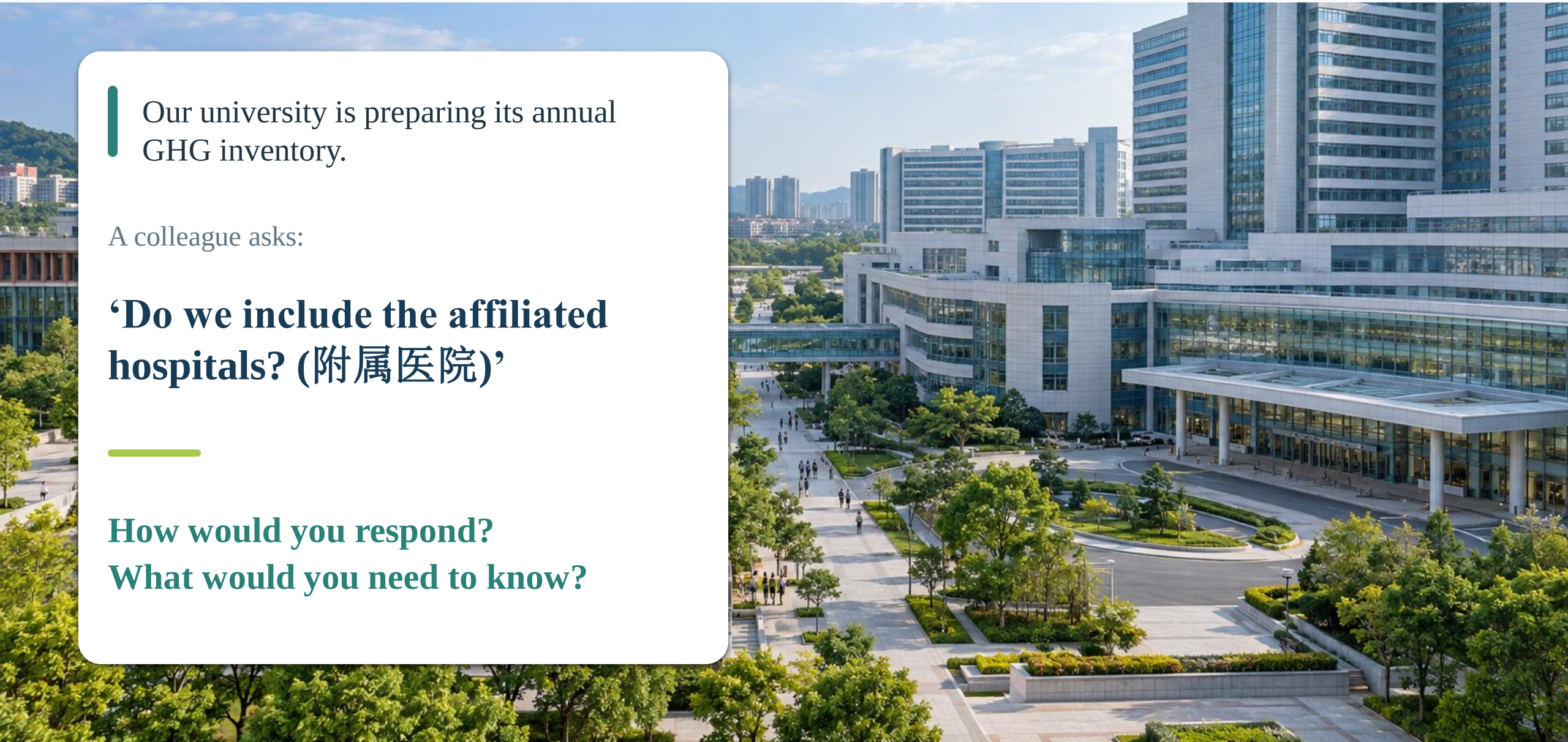
Financial Control: Venture B and Venture C

Equity Share:
- 80% of Venture A
- 30% of Venture B, and
- 100% of Venture C

Question:
which method should we choose?

Ans: it depends.

A question from our campus.



Our university is preparing its annual GHG inventory.

A colleague asks:

‘Do we include the affiliated hospitals? (附属医院)’

How would you respond?
What would you need to know?

What information changes our decision?

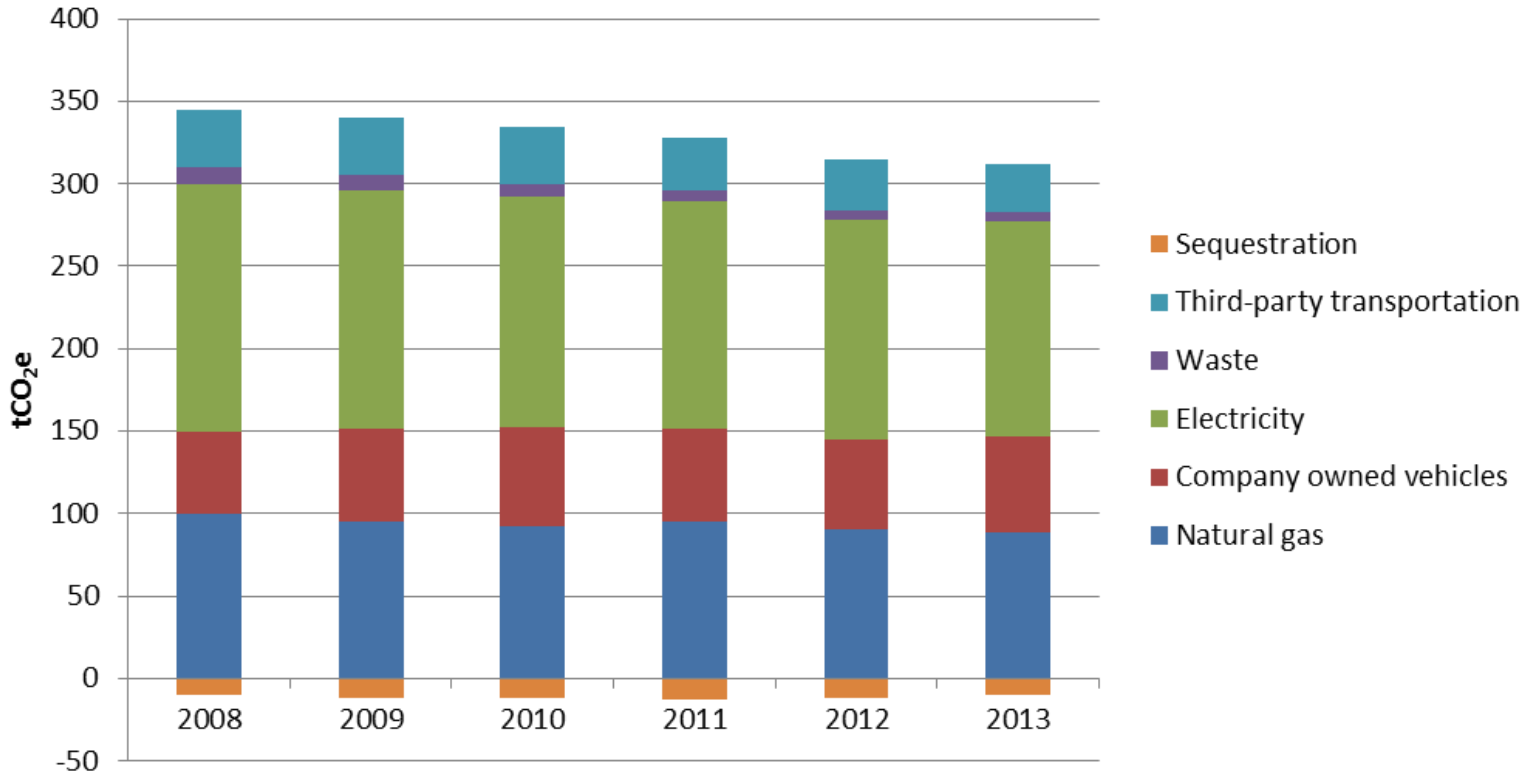
Reveal 1

The hospital is a separate public institution with its own legal identity.

Does this settle the question?

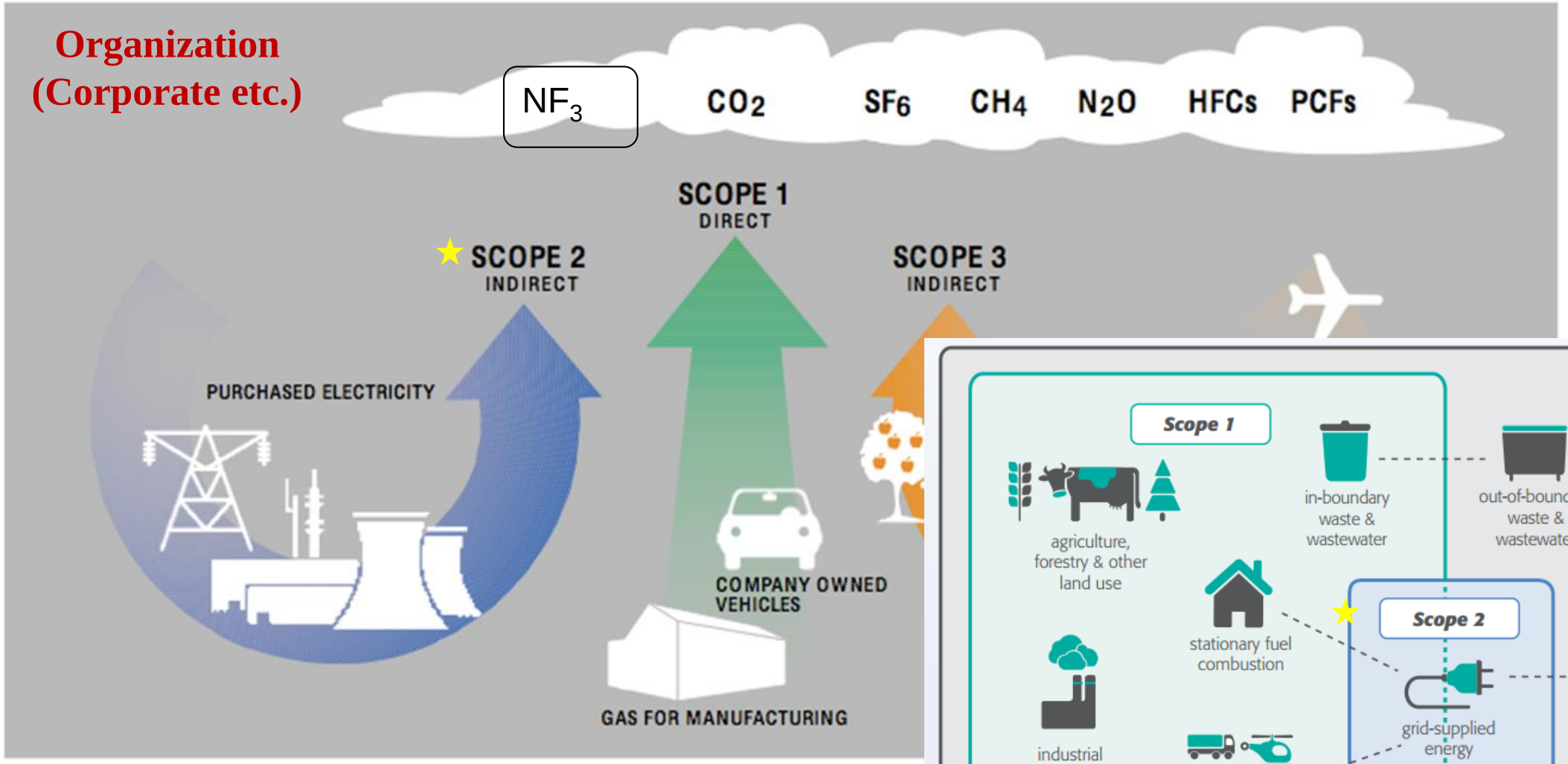


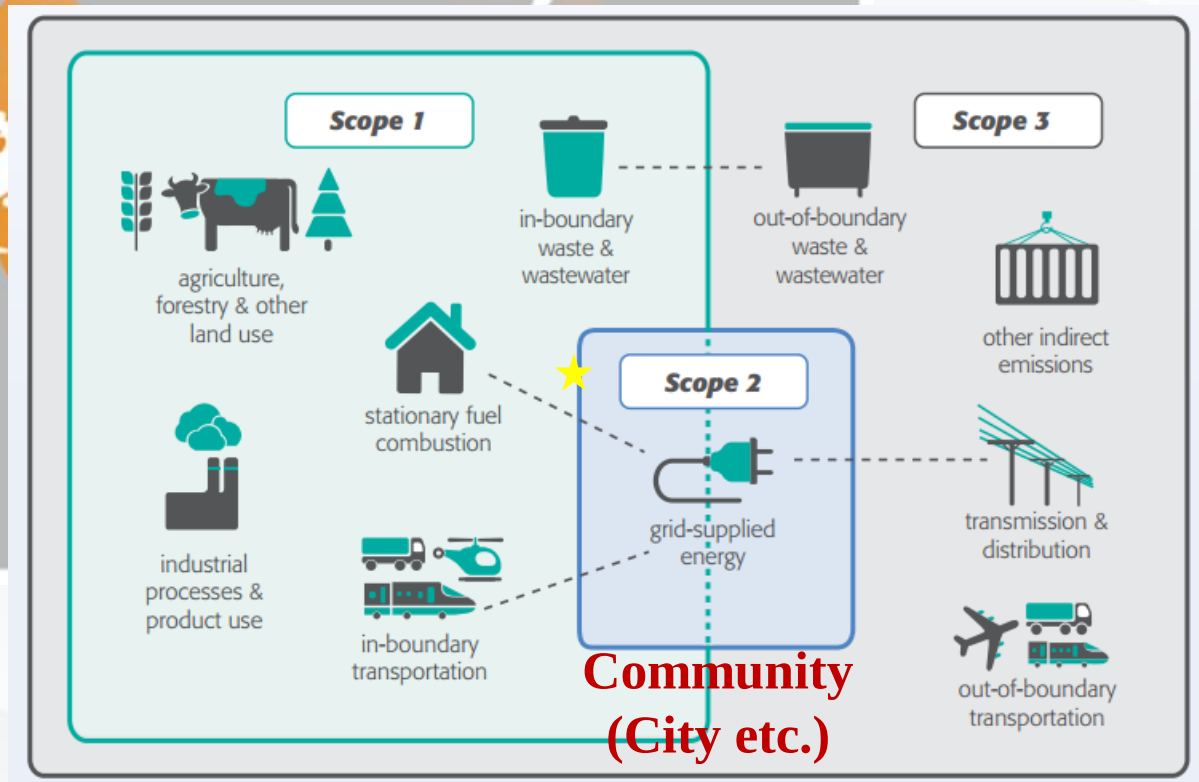
Step 2: Set operational boundary



For organizational-level GHG accounting, establishing an inventory boundary involves **identifying emissions** stemming from a defined organization's operations, **classifying** them into direct and indirect categories, and determining **the accounting and reporting scopes** for indirect emissions.

Organization (Corporate etc.)

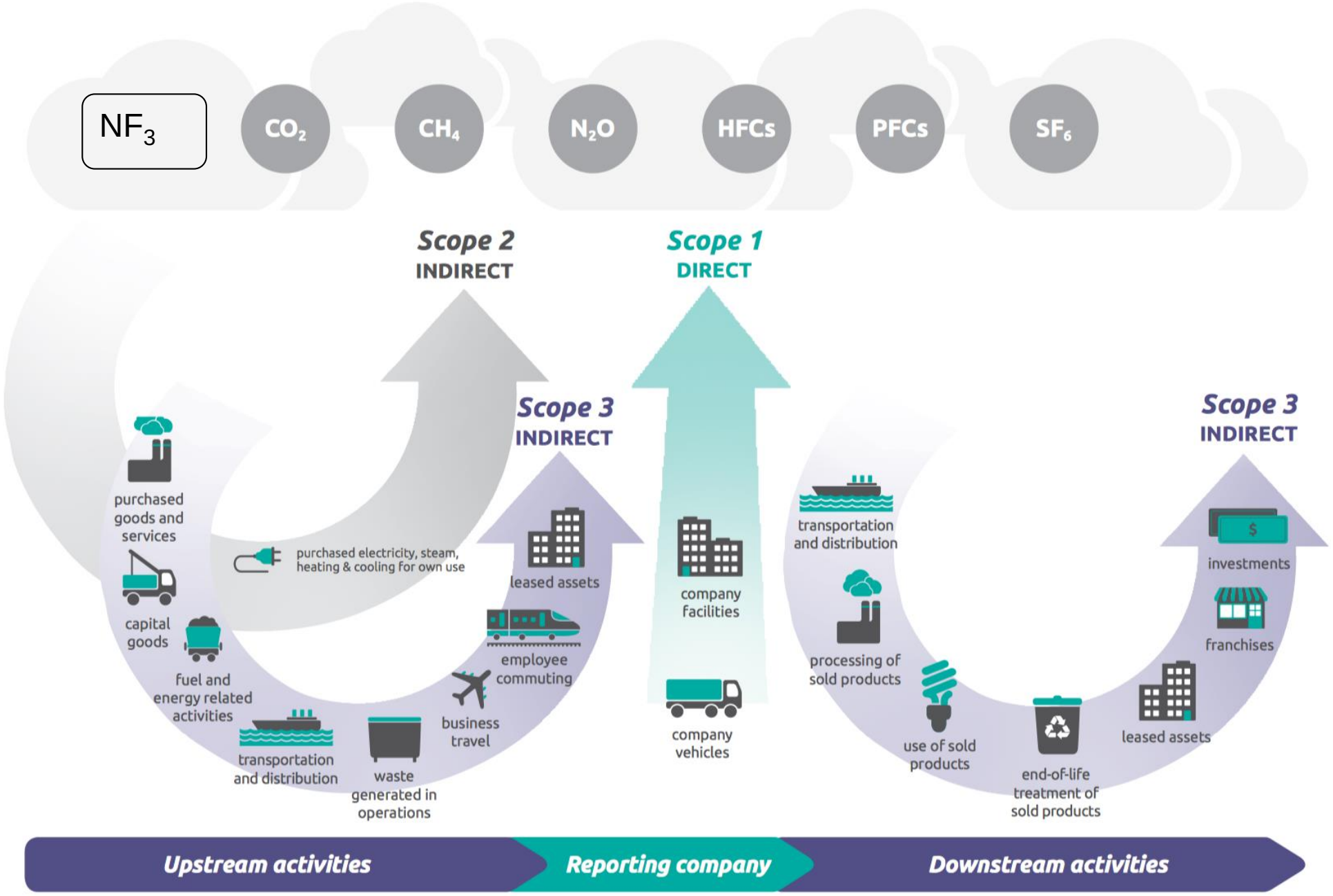




Community (City etc.)

★ **Note: the difference in the definition of "scope 2" in the two applications.**

Figure [1.1] Overview of GHG Protocol scopes and emissions across the value chain



Appendix A

Greenhouse gas emissions

Apple's corporate emissions

We account for our carbon footprint by following internationally recognized standards, like the World Resources Institute (WRI) Greenhouse Gas (GHG) Protocol and ISO 14040/14044.[1] Improving the accuracy of our carbon footprint is an ongoing process — as we learn more, we refine our carbon models and adjust our climate roadmap. We also regularly revisit the boundary of our carbon footprint as our data sources improve and our business evolves. For example, in 2022, we expanded our corporate footprint to include work from home emissions, third-party cloud services, electricity transmission and distribution losses, and upstream impacts from our scope 1 fuel use.

		Fiscal year				
		2022	2021	2020	2019	2018
Corporate emissions (metric tons CO_2e)[2]	**Gross emissions**	**324,100**	**166,380**	**334,430**	**573,730**	**586,170**
	Scope 1	**55,200**	**55,200**	**47,430**	**52,730**	**57,440**
	Natural gas, diesel, propane	39,700	40,070	39,340	40,910	42,840
	Fleet vehicles	12,600	12,090	4,270	6,950	11,110
	Other emissions[3]	2,900	3,040	3,830	4,870	3,490
	Scope 2 (market-based)[4]	**3,000**	**2,780**	**0**	**0**	**8,730**
	Electricity	0	0	0	0	8,730
	Steam, heating, and cooling[5]	3,000	2,780	0	0	0
	Scope 3	**265,800**	**108,400**	**287,000**	**521,000**	**520,000**
	Business travel 商务差旅	113,500	22,850	153,000	326,000	337,000
	Employee commute[6] 员工通勤	134,200	85,570	134,000	195,000	183,000
	Upstream fuel 上游燃料消耗	10,600	0	0	0	0
	Work from home (market-based) 居家办公	7,500	0	0	0	0
	Transmission and distribution loss (market-based) 电网输配电损失	0	N/A	N/A	N/A	N/A
	Third-party cloud (market-based) 第三方云存储	0	0	0	0	0
	Carbon removals					
	Corporate carbon offsets[7]	–324,100	–167,000[8]	–70,000[9]	0	0
Product life cycle emissions (metric tons CO_2e)[10]	**Gross emissions (Scope 3)**	**20,280,000**	**23,020,000**	**22,260,000**	**24,460,000**	**24,550,000**
	Manufacturing (purchased goods and services)	13,400,000	16,200,000	16,100,000	18,900,000	18,500,000
	Product transportation (upstream and downstream)	1,900,000	1,750,000	1,800,000	1,400,000	1,300,000
	Product use (use of sold products)	4,900,000	4,990,000	4,300,000	4,100,000	4,700,000
	End-of-life processing	80,000	80,000	60,000	60,000	50,000
	Carbon removals					
	Product carbon offsets	0	–500,000[11]	0	0	0
Total gross scope 3 emissions (corporate and product) (metric tons CO_2e)		**20,550,000**	**23,128,400**	**22,547,000**	**24,980,000**	**25,070,000**
Total gross carbon footprint (without offsets) (metric tons CO_2e)[12]		**20,600,000**	**23,200,000**	**22,600,000**	**25,100,000**	**25,200,000**
Total net carbon footprint (after applying offsets) (metric tons CO_2e)[12]		**20,300,000**	**22,530,000**	**22,530,000**	**25,100,000**	**25,200,000**

What Should Count for SJTU?

SJTU as an organization



SJTU
Annual organizational emissions

- **Electricity used in university buildings**
- **Electricity used in a university-affiliated hospital**
- **Electricity used in student dormitories**
- **Staff and student commuting**
- **University-funded business travel**
- **Fuel used by university vehicles**
- **Production of laboratory equipment**
- **Methane released from university wastewater treatment**
- **Food served in campus canteens**
- **…**

Which scope does each activity belong to? What kind of data do you need if you want to quantify the corresponding GHG emissions?

Step 3: Set the time period

- Typically calendar year (e.g. 2025) or financial year (e.g. 1 April 2025 to 31 March 2026)

- Need to consider:
  - Reporting requirements (e.g. alignment with financial reporting)
  - Data availability



Step 4: Collect activity data

- Electricity and gas meter readings (e.g. 20,000 kWh of grid electricity)
- Fuel invoices and financial data (e.g. 10,000 litres of diesel)
- Service records (e.g. refrigerant gas top-ups, waste collection)
- Floor area
- Surveys
- Proxy data (e.g. last year’s data)

…

表3　报告主体数据及来源

温室气体排放源	数据来源
固定燃烧源	企业能源平衡表
移动燃烧源	企业能源平衡表
过程排放源	原料消耗表 水平衡表（废水量） 废水监测报表（BOD、COD浓度） 财务报表（原料购买量/购买额）
逸散排放源	监测报表
购入电力、热力或蒸汽	企业能源平衡表 财务报表（相关销售额） 采购发票或凭证
生物燃料运输设备	企业能源平衡表 财务报表（生物燃料消耗量/运输货物重量、里程） 采购发票或凭证
固碳产品	产品产量表 财务报表（产值）
交通运输	财务报表（产值）
所使用的产品和服务隐含的碳排放	上游企业报告
所生产的产品和服务的碳排放	企业报告

参考：《工业企业温室气体排放核算和报告 通则》（2025）

附 录 B

（规范性）

非化石能源电力排放因子的取值要求及证明文件

B.1 电力排放因子取值

非化石能源电力排放因子取值要求如下。

a) 非化石能源电力直供企业使用且未并入市政电网的非化石能源电力消费量的排放因子为零。

b) 通过市场化交易购入使用的非化石能源电力消费量的排放因子可计为零或根据报告需求方要求单独进行核算与报告。

c) 全国电力平均二氧化碳排放因子(不包括市场化交易的非化石能源电量)应采用生态环境部和国家统计局发布的数据,如有更新,采用其最新发布的数值。

B.2 相关证明文件

应提供发电与用电双方签订的市场化交易合同,以及按合同执行的绿色电力证书交易凭证和由省级及以上电力交易机构出具的交易结算凭证。交易结算凭证应载明在核算与报告周期内的月度结算电量及其项目类型、发电企业名称、用电企业名称等。绿色电力证书交易凭证载明的内容应包括项目名称、项目代码、项目类型、项目所在地、电量生产日期等。

直供企业使用且未并入市政电网的非化石能源电力消费量应提供每月电量统计原始记录。





国家温室气体排放因子库

生态环境部
国家统计局

公告

公告 2025年 第47号

关于发布2023年电力二氧化碳排放因子的公告

生态环境部、国家统计局组织计算了2023年全国、区域和省级电力平均二氧化碳排放因子，全国电力平均二氧化碳排放因子（不包括市场化交易的非化石能源电量），以及全国化石能源电力二氧化碳排放因子，供核算电力消费的二氧化碳排放量时参考使用。上述因子与《关于发布2021年电力二氧化碳排放因子的公告》（公告2024年第12号）、《关于发布2022年电力二氧化碳排放因子的公告》（公告2024年第33号）中2021年、2022年电力二氧化碳排放因子的计算方法和数据来源一致。

特此公告。

附件：2023年电力二氧化碳排放因子

生态环境部

国家统计局

2025年12月31日

生态环境部办公厅2025年12月31日印发

解读：生态环境部应对气候变化司有关负责人就《关于发布2023年电力二氧化碳排放因子的公告》答记者问

表 1 2023 年全国电力平均二氧化碳排放因子

	因子（$kgCO_2/kWh$）
全国	0.5306

表 4 2023 年全国电力平均二氧化碳排放因子（不包括市场化交易的非化石能源电量）

	因子（$kgCO_2/kWh$）
全国	0.6096

表 5 2023 年全国化石能源电力二氧化碳排放因子

	因子（$kgCO_2/kWh$）
全国	0.8273

- Geographical appropriateness
- Temporal (time) appropriateness
  - Publication date, and validity date
- Technological appropriateness

- BEIS's emission factors (applied to UK firms ONLY):
  - https://www.gov.uk/government/collections/government-conversion-factors-for-company-reporting
- China's National GHG Emission Factor Database (version 2 published in March 2026, with 576 emission factors) https://data.ncsc.org.cn/factories/index
- IPCC; US EPA; IEA; Ecoinvent; + many more

Step 5: Apply emission factors and convert to CO_2e

Activity Data		**Emission factor**		**Global warming potential**		**kgCO_2e**
20,000 kWh of electricity (in UK)	*	0.45844 kgCO_2/kWh	*	1	=	9,169
		0.000017 kgCH_4/kWh		28		9
		0.000011 kgN_2O/kWh		265		57
Total						9,235

Step 6: Reporting: examples of summary table

	tCO_2	tCH_4	tN_20	tCO_2e
Scope 1				
Company owned vehicles	2,370	2	0.2	2,480
Scope 2				
Electricity consumption	154	0.5	0.01	169
Scope 3				
Staff commuting	298	1	0.2	383
Waste disposal	0	10	0	250
Total	**2,822**	**14**	**0.4**	**3,282**
Biogenic CO_2	26	0	0	26
Total incl biogenic emissions	**2,848**	**14**	**0.4**	**3,308**

M&S
EST. 1884

M&S Group CO_2e emissions

	This year 2019/20 000t CO_2e	Last year 2018/19 000t CO_2e	Baseline 2006/07 000t CO_2e	% change on 2006/07
Carbon offsets purchased and retired	195	158	0	–
Total net emissions	**0**	**0**	**640**	**–**
Total net emissions per 1,000 sq ft of salesfloor	**0**	**0**	**40**	**–**

M&S global Location-based (gross) emissions

000 tonnes CO2e	2006/07	2015/16	2016/17	2017/18	2018/19	2019/20
Store & office gas heating	39	34	37	39	38	36
Store & office fridge & air con	130	61	58	55	42	48
Warehouse gas heating	9	7	6	6	5	5
				1	3	2
				80	80	81
				181	167	173
				223	175	150
				25	18	16
				248	193	165
				430	360	338
				23	19	18
				23	17	14
				2	2	2
				14	15	1
				2	2	1
				41	35	33
				471	395	370
				25	21	19

Mostly grid-average emission factor

M&S global Market-based emissions

000 tonnes CO2e	2006/07	2015/16	2016/17	2017/18	2018/19	2019/20
Store & office gas heating	39	28	26	14	29	36
Store & office fridge & air con	130	61	58	55	42	48
Warehouse gas heating	9	7	6	6	5	5
Warehouse fridge & air con	6	2	2	1	3	2
Delivery transport fuel	62	78	82	80	80	81
(Direct) Scope 1	246	176	173	157	158	173
Store & office electricity	360	22	0	0	0	11
Warehouse electricity	34	13	0	0	0	11
(In-direct) Scope 2	394	35	0	0	0	22
Total Scopes 1& 2	639	211	173	157	158	195
Index /1000 sq ft of sales floor	40	11	9	8	8	10
Electricity T&D	34	27	27	23	17	14
Fuel use T&D (from 2011)	0	2	2	2	2	2
Business travel	18	24	17	14	15	15
Waste	40	2	2	2	2	1
(Other) Scope 3	92	55	48	41	35	33
Total Scopes 1, 2 &3	732	266	221	198	193	228
Index /1000 sq ft of sales floor	46	14	11	10	10	12

Dependent on contractual instruments

E.g., RECs （绿证）

Introduction to China's Standards

Org. boundary: legal entity

Direct emissions + Indirect emissions

GHG gases covered：CO_2，CH_4，N_2O，HFCs，PFCs，SF_6，NF_3

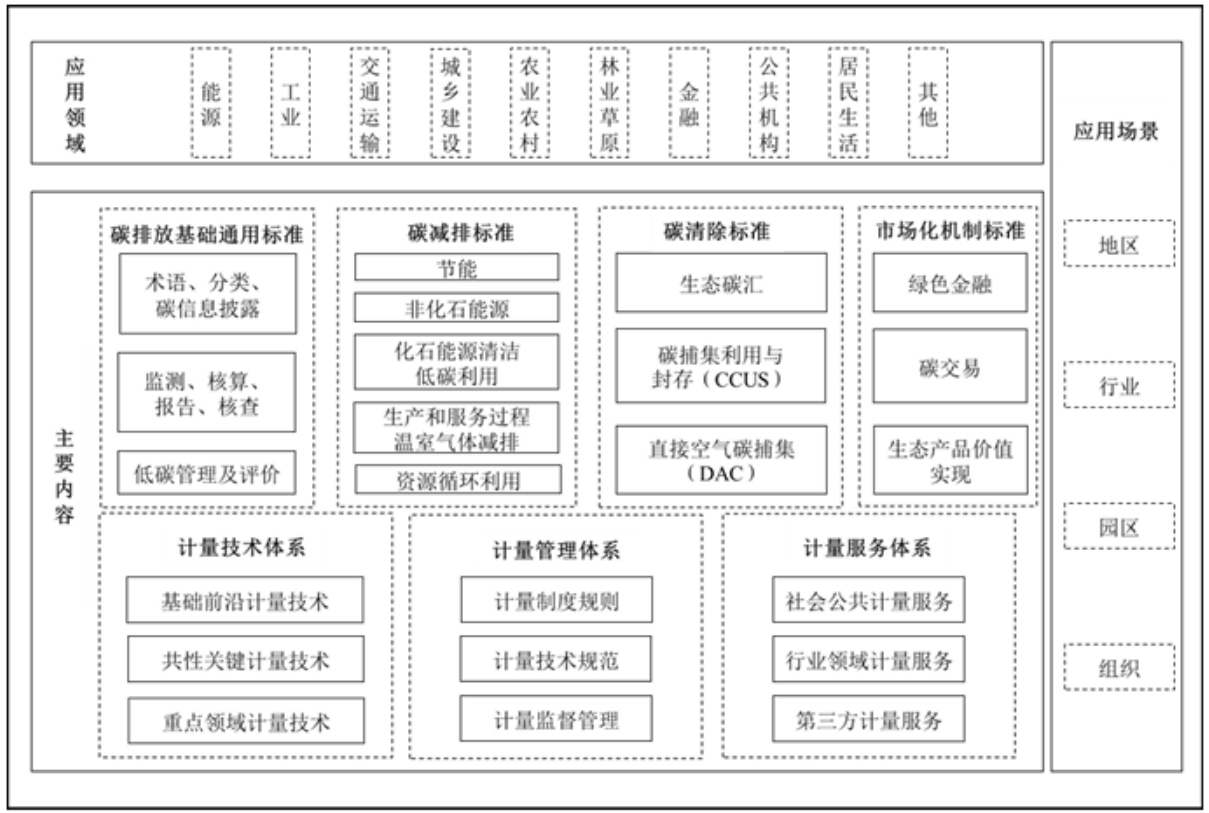


图 1 碳达峰碳中和标准计量体系框架图

来自：《建立健全碳达峰碳中和标准计量体系实施方案》

GB/T 32151 《温室气体排放核算与报告要求》从不同的企业层面规定了温室气体核算与报告的要求。根据“全国标准信息公关服务平台”的最新信息，目前GB/T 32151已经发布了50+部分，覆盖多个行业

上海节能低碳和应对气候变化网 (reg-sh.org) (unavailable at this moment)

上海市温室气体排放核算与报告指南（试行） （沪发改环资〔2012〕180 号）

上海市电力、热力生产业温室气体排放核算与报告方法（试行） （沪发改环资〔2012〕181 号）

钢铁行业

化工行业

有色金属行业

纺织和造纸行业

非金属矿物制品业

航空运输业

旅游饭店、商场、房地产业及金融业

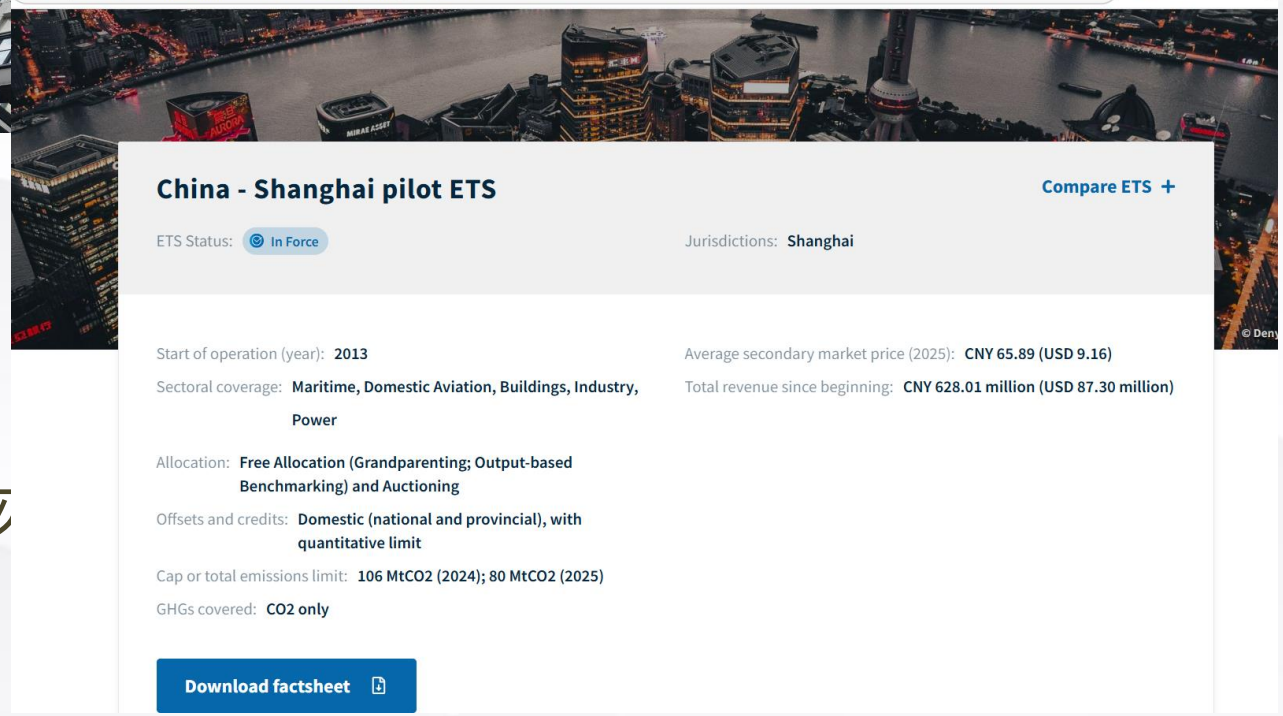
运输站点行业

水运业



In Shanghai





Shanghai: GHG accounting and reporting guidelines

- 上海市温室气体排放核算于报告指南（试行）（沪发改环资【2012】180号
- 上海市电力、热力生产业温室气体排放核算于报告指南（试行）（沪发改环资【2012】181号
- 上海市钢铁行业温室气体排放核算于报告指南（试行）（沪发改环资【2012】182号
- 上海市化工行业温室气体排放核算于报告指南（试行）（沪发改环资【2012】183号
- 上海市有色金属行业温室气体排放核算于报告指南（试行）（沪发改环资【2012】184号
- 上海市纺织、造纸行业温室气体排放核算于报告指南（试行）（沪发改环资【2012】185号
- 上海市非金属矿物制品业温室气体排放核算于报告指南（试行）（沪发改环资【2012】186号
- 上海市航空运输业温室气体排放核算于报告指南（试行）（沪发改环资【2012】187号
- 上海市旅游饭店、商场、房地产业及金融业办公建筑温室气体排放核算于报告指南（试行）（沪发改环资【2012】188号
- 上海市运输站点行业温室气体排放核算于报告指南（试行）（沪发改环资【2012】189号
- 上海市水运行业温室气体排放核算于报告指南（试行）（沪发改环资【2016】10号

If an enterprise based in Shanghai needs to prepare a greenhouse gas (GHG) emissions report, which guidelines should it follow?



Quick exercise



Four activities on our campus.

University X uses operational control to consolidate GHG inventory. Assume that only four activities are relevant.



Backup diesel generator



University-owned petrol vehicles



Purchased grid electricity



Faculty business flights

Apply the scope rules.

1. **Identify the scope.**
2. **Explain the source–university relationship.**
3. **Name the annual activity data and unit needed.**

Activity	Scope	Reason	Activity data + unit
Backup generator	**Scope 1**	University-controlled diesel combustion	Litres of diesel consumed per year.
University petrol vehicles			
Purchased grid electricity			
Faculty business flights			

University X's data in 2025

Source	Annual activity	Teaching factor	Emissions in kg CO_2e
Backup generator	200 L diesel consumed	2.70 kg CO_2e/L	
Petrol vehicles	300 L petrol consumed	2.30 kg CO_2e/L	
Grid electricity	20,000 kWh consumed	0.50 kg CO_2e/kWh	
Faculty business flights	8,000 passenger-km	0.12 kg CO_2e/passenger-km	

Note: All numbers and factors are invented for practice and are not real university data. Factors already include CO_2e.

Summary Table for University X's GHG Inventory in 2025				
Category	Scope 1	Scope 2	Scope 3	Total
Results	_____ kg	_____ kg	_____ kg	_____ kg (= ___ t)

Some of the Challenges in organization level accounting (*)

Some of the challenges

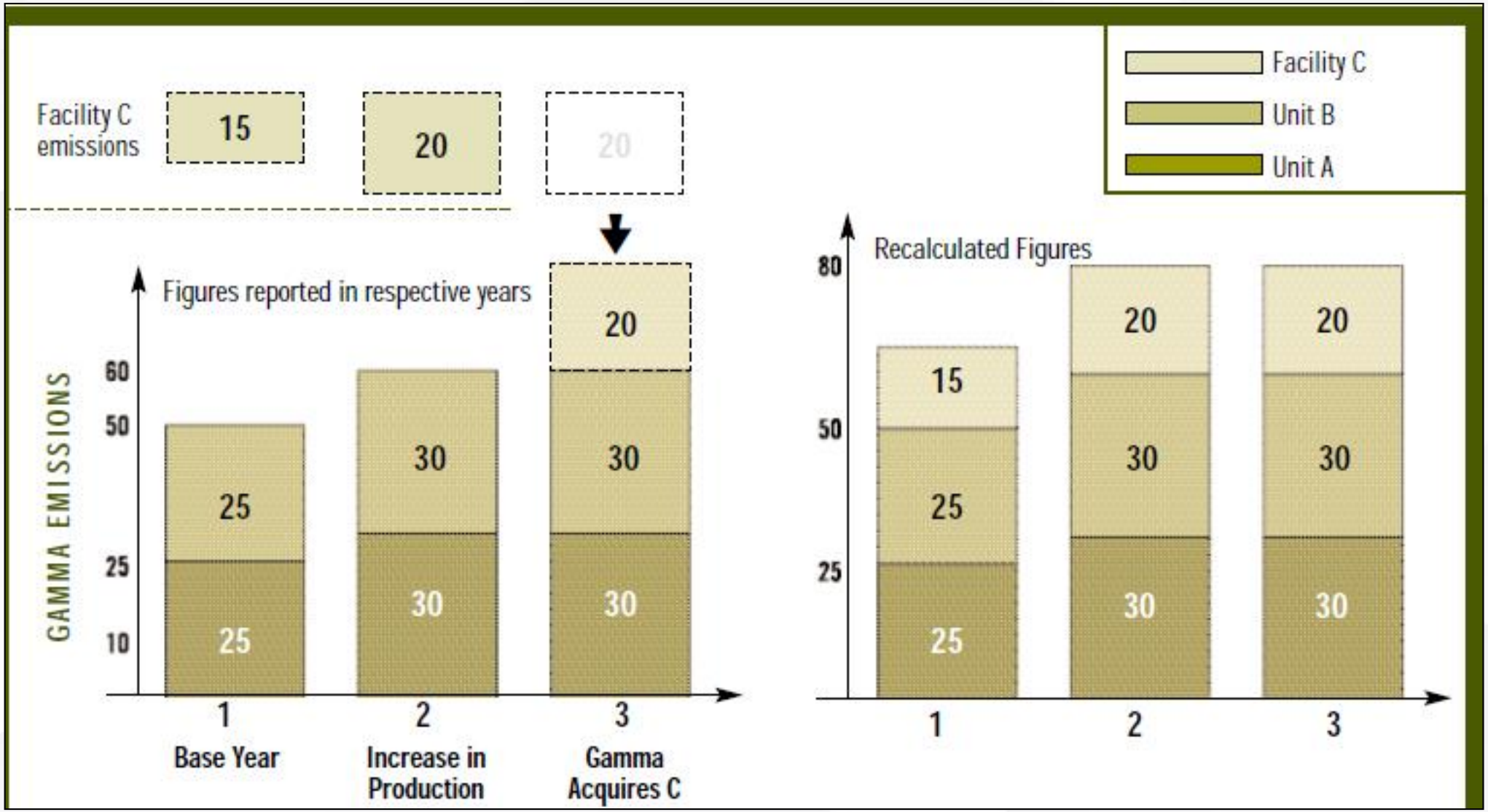
- Base year recalculations
- Biogenic emissions
- Leased asset (*)
- Combined heat and power (CHP) (*)
- Contractual emission factors – scope 2 (*)



Base year recalculation

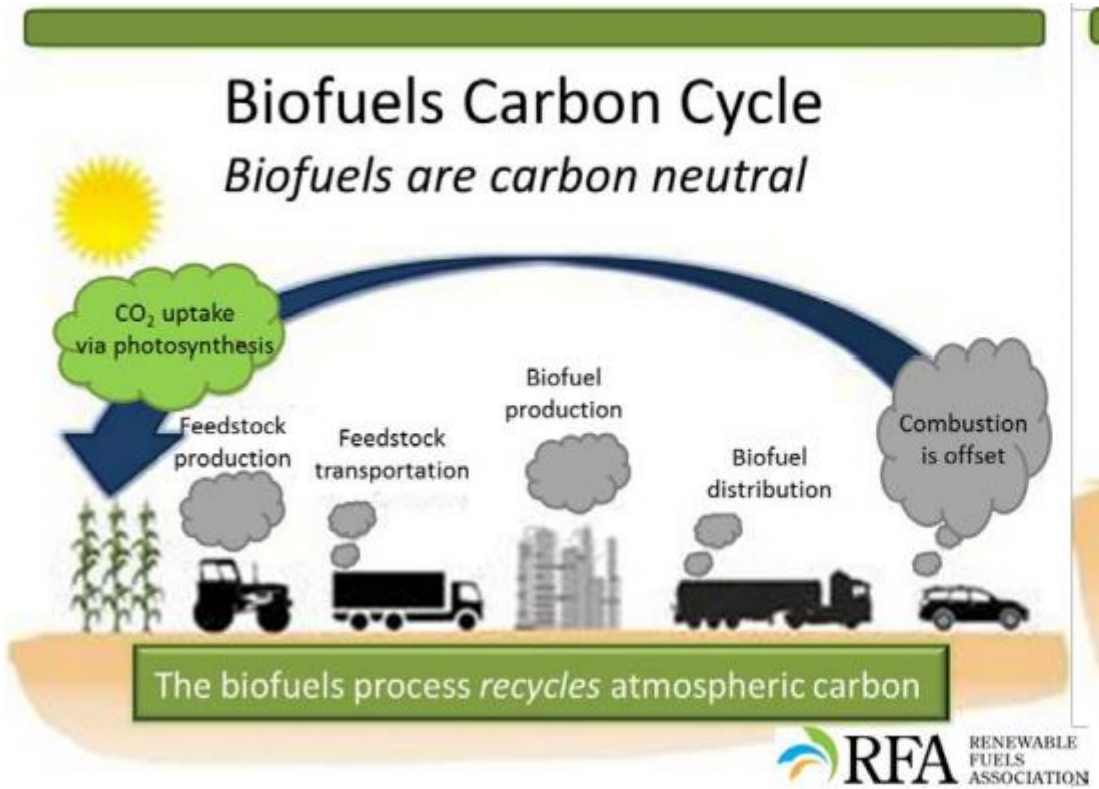
- Base year is the reference period for tracking changes in emissions over time

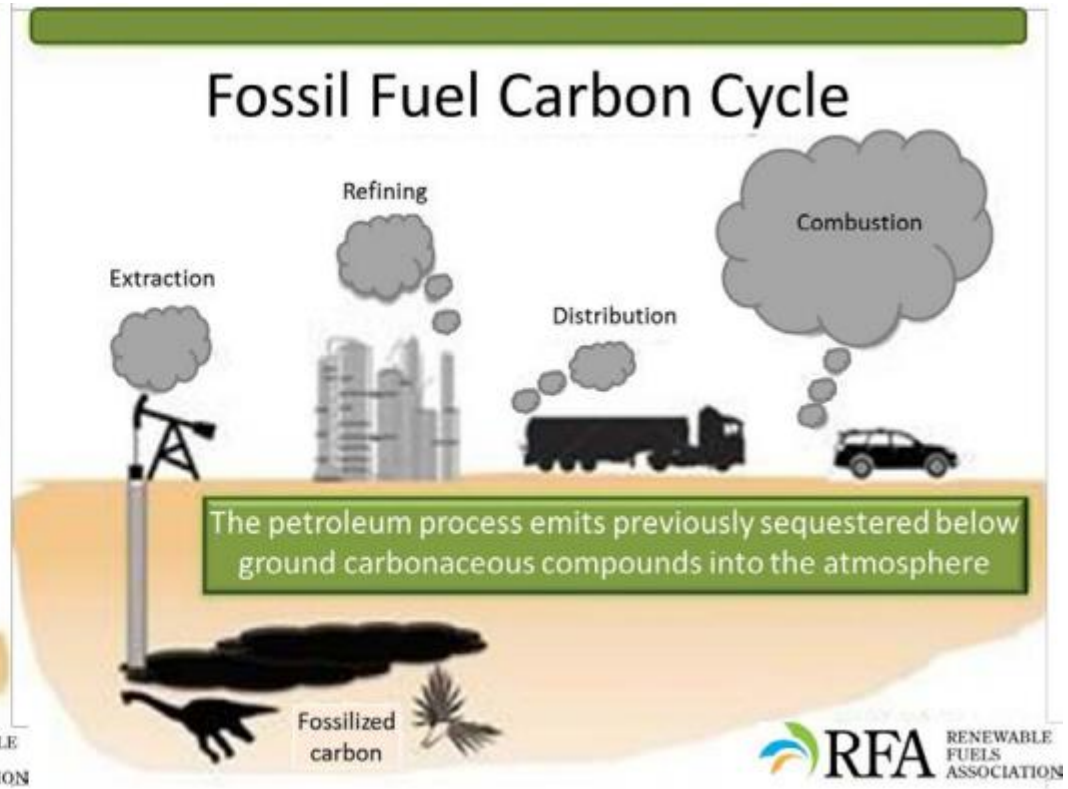
- Recalculation may be needed due to:
  - Mergers, acquisitions or divestments
  - Outsourcing or insourcing operations (between scopes 1 and 3)
  - Changes in calculation methodology
  - Discovery of significant errors


Facility C emissions
15
20
20
Facility C
Unit B
Unit A
Figures reported in respective years
GAMMA EMISSIONS
60
50
25
10
25
25
30
30
30
30
20
1
2
3
Base Year
Increase in Production
Gamma Acquires C
Recalculated Figures
80
50
25
15
25
25
20
30
30
20
30
30
1
2
3

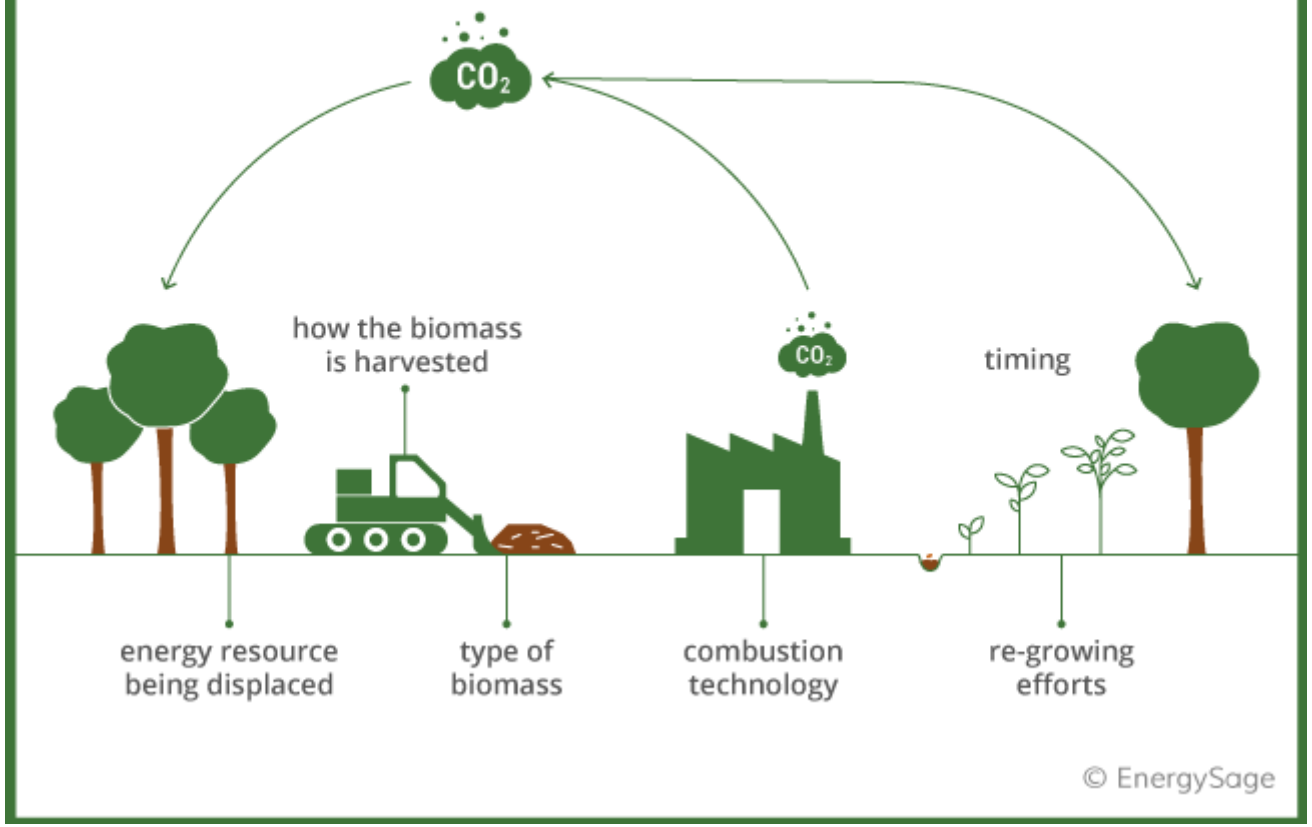
Biogenic emissions

- Report emissions separately from scopes 1, 2, and 3. → Why ?





Factors that contribute to carbon-neutral bioenergy



Wrap up – Organizational GHG inventory

Why – why firms bother with GHG inventory?

Internal

External

What – what is counted?

A defined organizational boundary

A defined operational boundary

Gases, Reporting year

How – how is it quantified?

Activity Data * Emission Factors

Organizational carbon accounting is about:

1. Set organizational boundary
2. Set operational boundary
3. Define time period, e.g. calendar year
4. Collect activity data
5. Apply emission factors and convert to CO_2e

6. Reporting





See you next time

What Should Count for SJTU?

SJTU as an organization



SJTU
Annual organizational emissions

- **Electricity used in university buildings**
- **Electricity used in a university-affiliated hospital**
- **Electricity used in student dormitories**
- **Staff and student commuting**
- **University-funded business travel**
- **Fuel used by university vehicles**
- **Production of laboratory equipment**
- **Methane released from university wastewater treatment**
- **Food served in campus canteens**
- **…**

Which scope does each activity belong to? What kind of data do you need if you want to quantify the corresponding GHG emissions?

Lecture 8

目 录

Contents

01- BMW’s Corporate Inventory and Decarbonization

02- Wrap up and Next time

* In-class Project activity

03

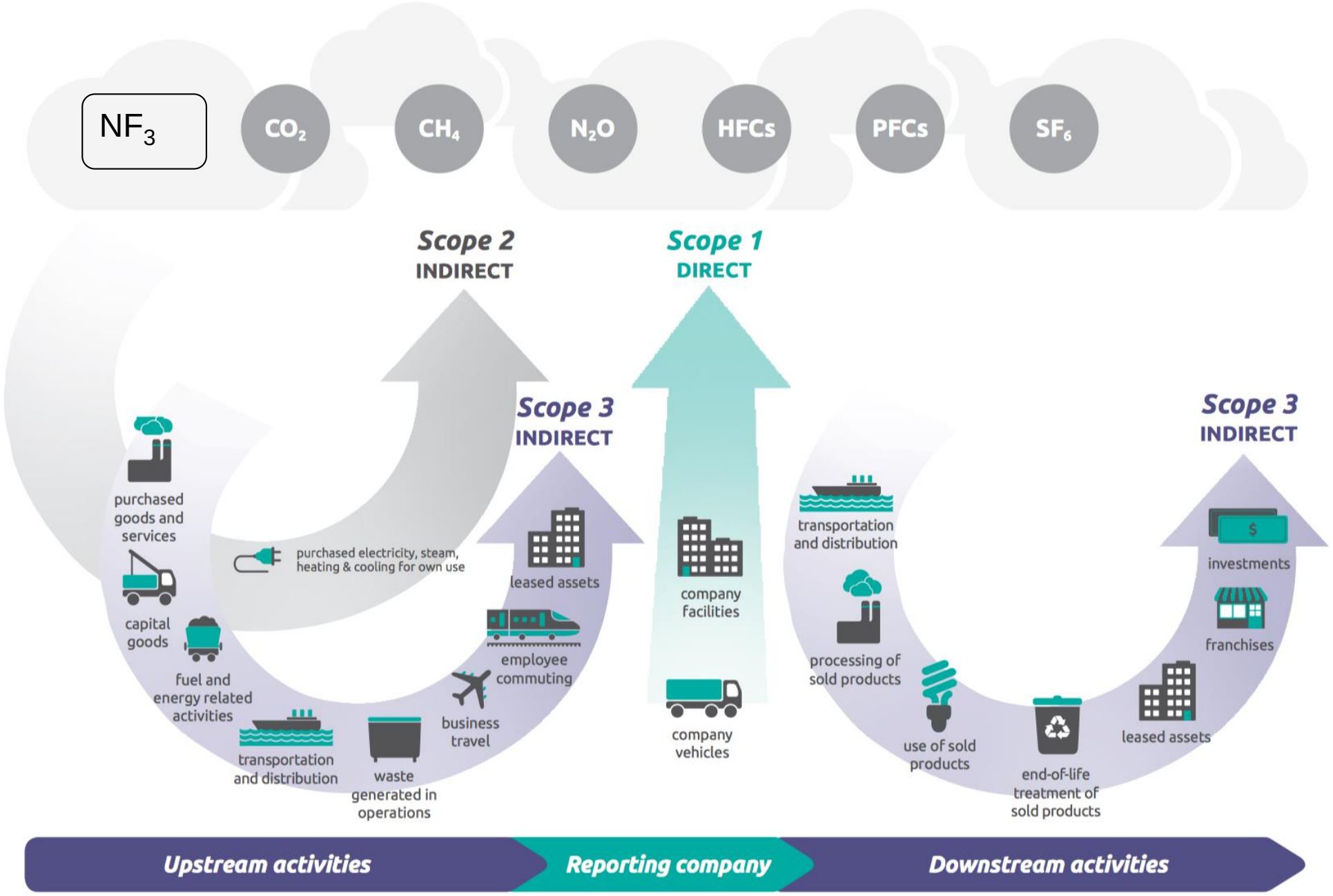
BMW’s Corporate Inventory and Decarbonization

How Many Parts Are in a Car?

Posted by Collectors Auto Supply on May 5th 2020

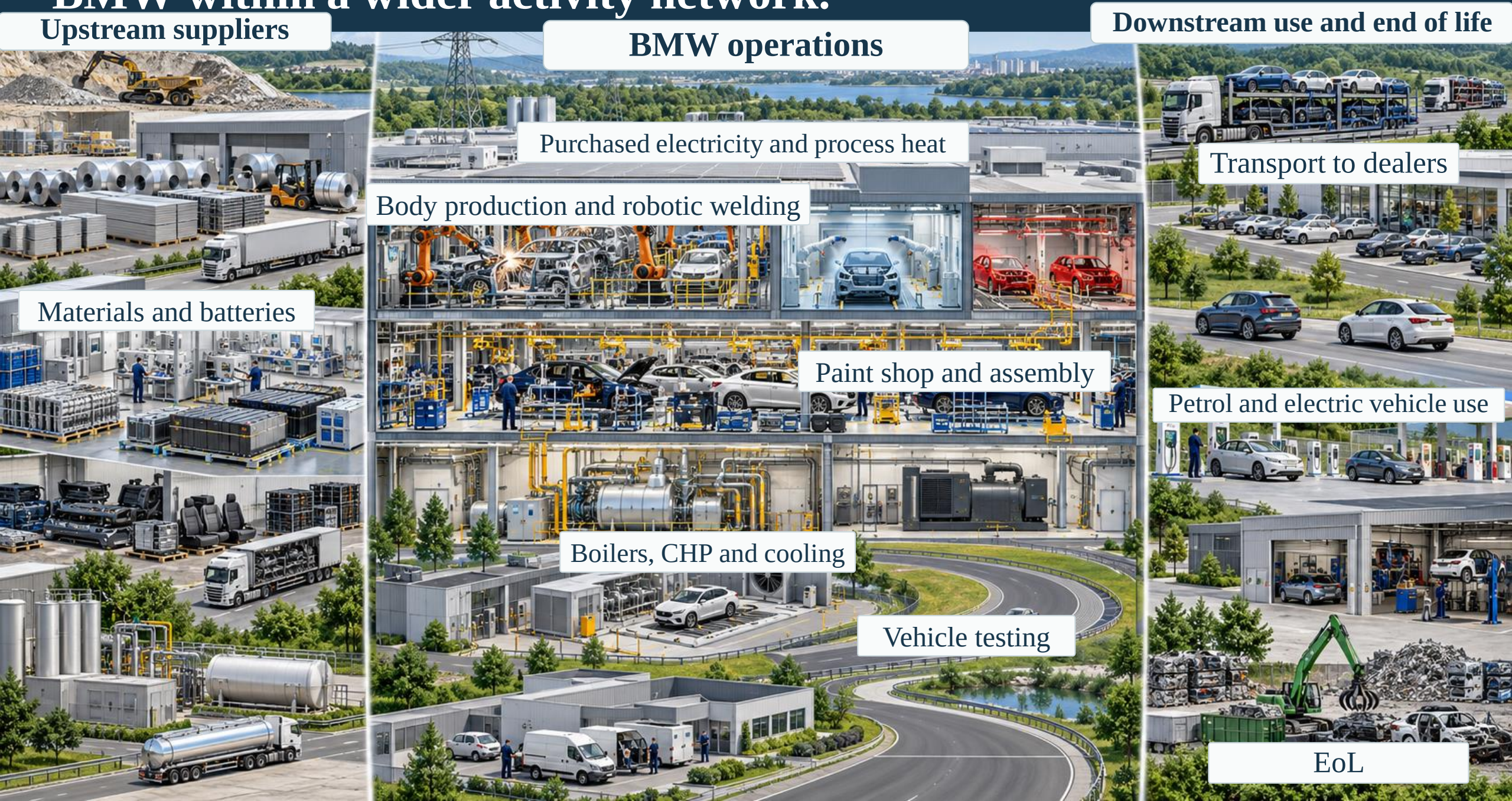
The average car has roughly 30,000 parts if you count all parts down to the nuts and bolts. If you consider large parts such as the engine as one part, you can say that a car has about 1,800 separate parts. The engine alone has thousands of individual parts inside it. Many parts are made by the vehicle manufacturer, but each manufacturer also has a number of suppliers that make parts to go into different cars.

Figure [1.1] Overview of GHG Protocol scopes and emissions across the value chain



BMW within a wider activity network.

Picture created by ChatGPT-5.6 Sol



Activity 1: Mapping BMW's emissions

See handed-out Task Sheet

Read BMW Corporate Inventory

- Different carbon emissions (scope 1 vs 2 vs 3) exhibit different characteristics
  - Difficulty to measure
  - Ability to control or influence
  - Business relevance
  - Scope 1/2/3 proportion varies by industry (ex: aviation vs. railroads)

Exhibit 11 Corporate-Wide Carbon Emissions of BMW Group, Mercedes-Benz, and Audi, 2019-2021

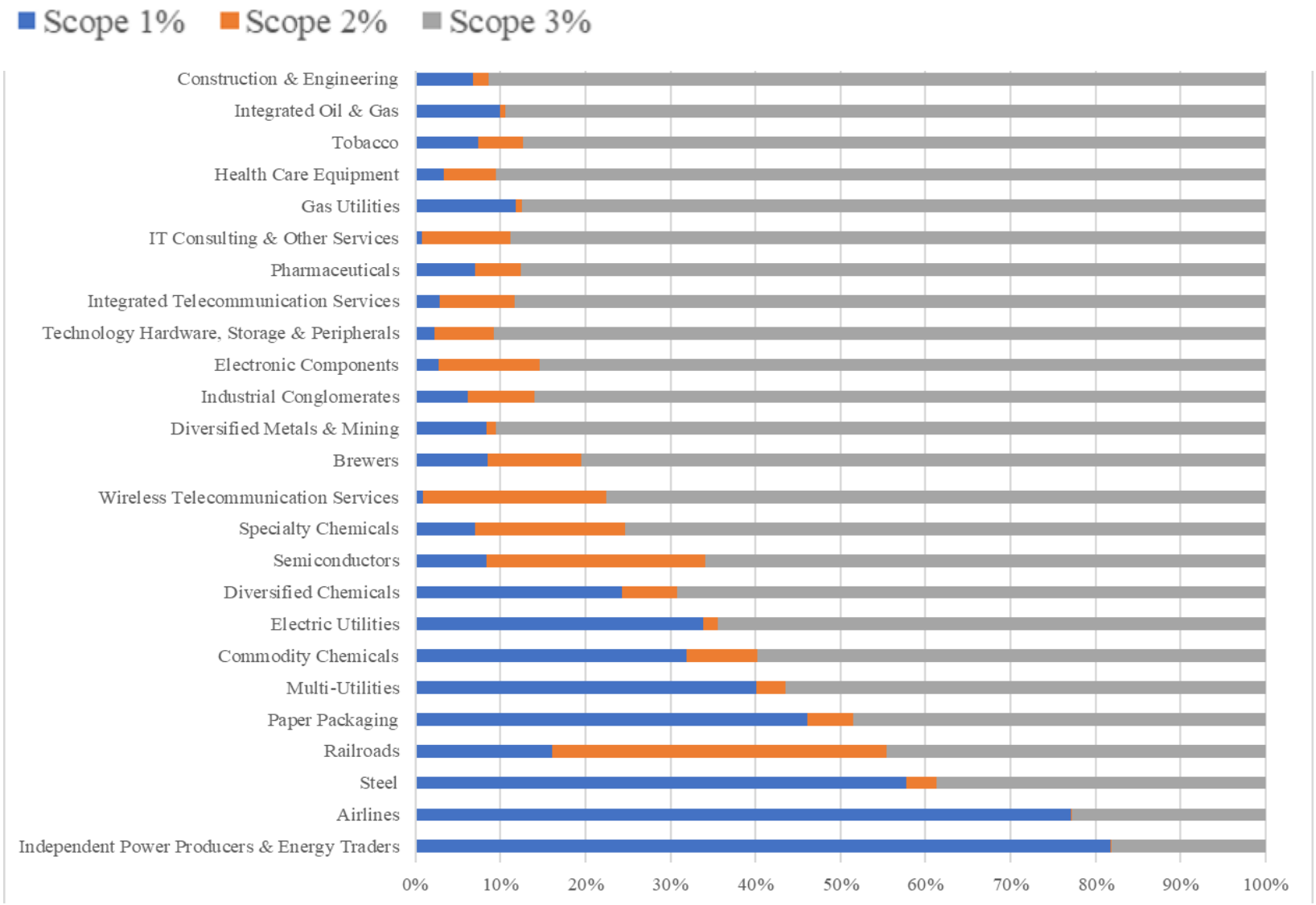
CO_2 Emissions by Scope and Brand, in million metric tons of CO_2e

	BMW			Mercedes-Benz			Audi		
	2021	2020	2019	2021	2020	2019	2021	2020	2019
Total emissions	122.54	118.49	133.55	124.29	NA	NA	NA	NA	NA
Scope 1	0.70	0.68	0.68	0.68	1.03	1.24	0.17	0.17	0.20
Scope 2	0.13	0.13	0.35	0.47	1.04	1.28	0.06	0.06	0.25
Scope 3	121.71	117.68	132.52	123.14	NA	NA	NA	NA	NA
Supply chain	18.53	16.23	18.51	20.40	NA	NA	NA	NA	NA
Use phase	99.81	98.78	110.90	99.00	NA	NA	NA	NA	NA
Other	3.37	2.67	3.11	3.74	NA	NA	NA	NA	NA
Deliveries (million vehicles)	2.52	2.33	2.54	2.33	2.46	2.82	1.69	1.70	1.85
Share of EVs (%)	13.0	8.3	5.8	9.8	7.4	2.0	NA	NA	NA
EU CO_2 emissions (in g / km)	115.9	99.1	127	115	104	137	122.1	102.9	130.6

Source: *BMW Group Report 2021*, https://www.bmwgroup.com/content/dam/grpw/websites/bmwgroup_com/ir/downloads/en/2022/bericht/BMW-Group-Report-2021-en.pdf, "Mercedes-Benz Group ESG Conference 2022 Presentation," https://group.mercedes-benz.com/dokumente/investoren/praesentationen/mercedes-benz-ir-esg-conference-2022-presentation.pdf, *Audi Report 2021*, https://www.audi.com/content/dam/gbp2/en/company/investor-relations/reports-and-key-figures/annual-reports/audi-report-2021.pdf, all accessed July 2022.

Notes: *Share of EVs* refers to the proportion of the company's total deliveries in a given year that were EVs, based on vehicle volume.
EU CO_2 emissions refers to the average emissions per kilometer that result from driving the company's new vehicle fleet sold in EU that year. This average is weighted by the volume of vehicles sold across each of the company's models. The testing procedure was updated from the New European Driving Cycle (NEDC) to the Worldwide Harmonised Light Vehicles Test Procedure (WLTP) starting in 2021.

GHG emissions by scope, by industry (select industries)



Source: Excerpted from George Serafeim, "Risks and Opportunities from the Transition to a Low Carbon Economy: A Business Analysis Framework." HBS No. 123-014, December 2022.

Question: Among the various scopes of emissions for BMW, which metric would you trust the most in making decisions?

Scope	Reliability	Comparability
Scope 1 and 2	**High**, measured for 20 years with reliable inputs	**High**, e.g., Scope 2 only contains electricity usage and hence more comparable than scope 3, which can include different components from different firms
Scope 3 Supply Chain	**Low**, most are not primary data but measured using averages from Gabi	**Low**, different firms use different amount of primary data
Scope 3 Use-Phase	**Medium**, not actual emissions but anticipated based on assumptions (e.g., 200 000 mileage)	**High**, follow standards like the WLTP and EPA

- **Reliability** captures how objective and verifiable the information is.
- **Comparability** captures how useful it is to compare the information across different companies or over time.

Source: Teaching Note – Driving Decarbonization at BMW, 5-123-049

Question: What institutions can help enhance the desirable properties of emissions measures?

	Reliability	Comparability
Institutions		

Source: Teaching Note – Driving Decarbonization at BMW, 5-123-049

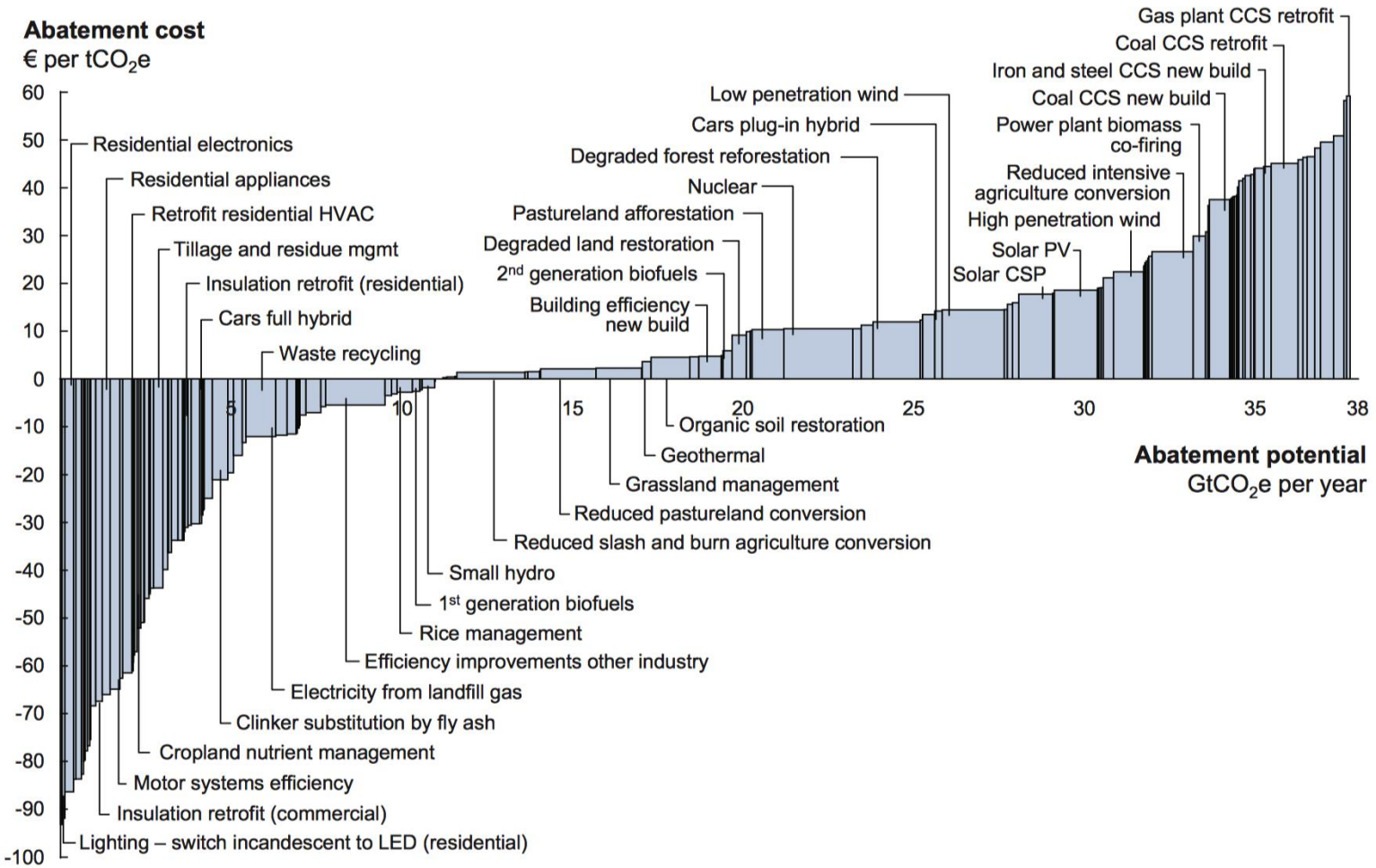
BMW: Decarbonization Strategy



Cost savings

Exhibit 1

Global GHG abatement cost curve beyond business-as-usual – 2030



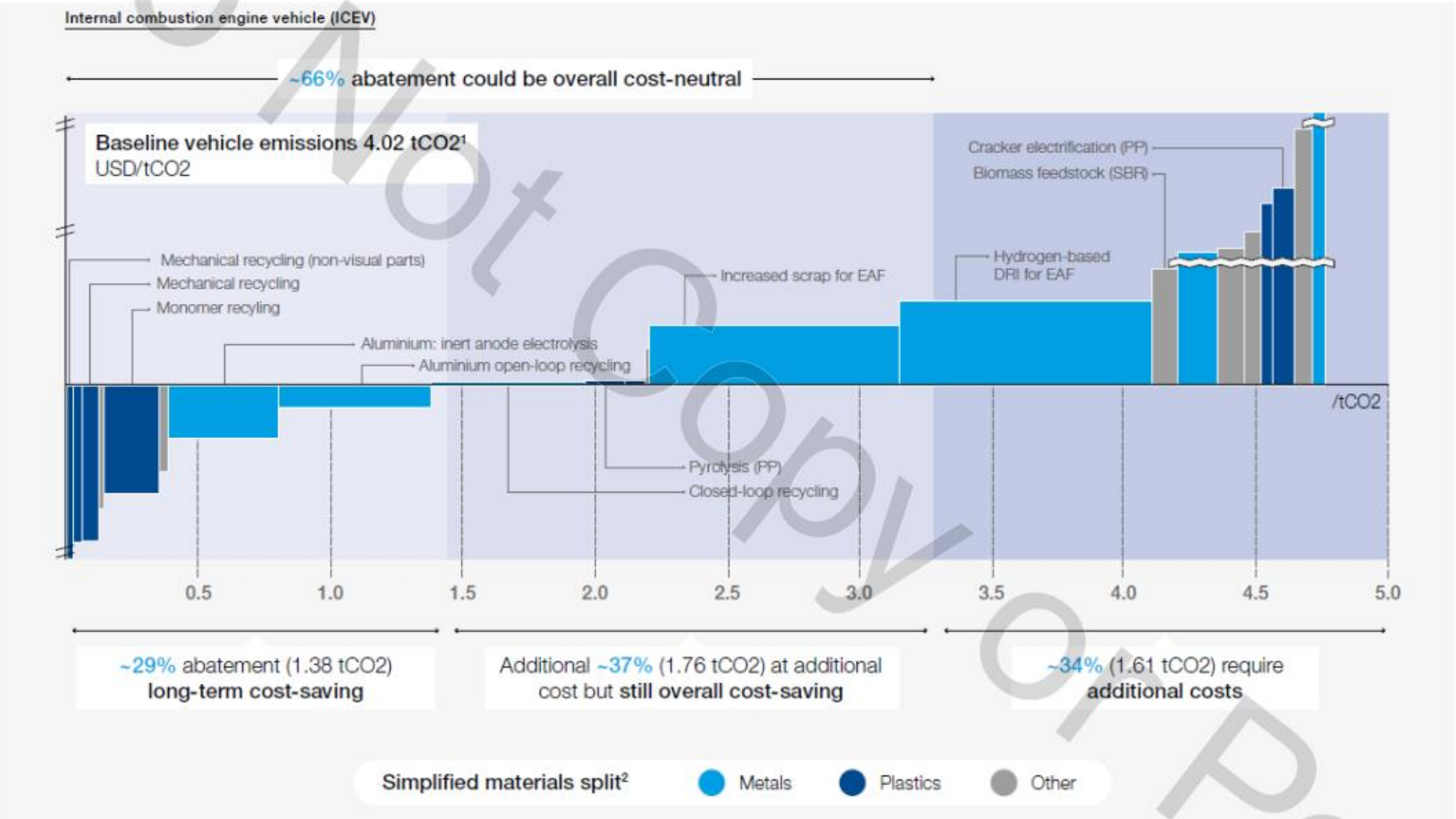
Note: The curve presents an estimate of the maximum potential of all technical GHG abatement measures below €60 per tCO_2e if each lever was pursued aggressively. It is not a forecast of what role different abatement measures and technologies will play.
Source: Global GHG Abatement Cost Curve v2.0

(Source: Mckinsey & Company

https://www.mckinsey.com/business-functions/sustainability/our-insights/greenhouse-gas-abatement-cost-curves)

BMW Marginal Abatement Cost Curve (for ICEV)

Exhibit 13 Carbon Abatement Cost Curve



1. In this analysis, a premium C-segment vehicle with 1.95t vehicle weight: 1.04t steel; 0.29t aluminium; 0.10t rubber; 0.07t PP; 0.03t PE; 0.05t glass is considered
2. Metals including steel, high-strength steel, aluminium, alumina; plastics including polypropylene, polyethylene, polyamide 6; other materials including rubber, glass

Source: McKinsey analysis (Team, McKinsey Decarbonization Pathways Optimizer)

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04

Wrap up and Next Time

Wrap up – understanding a city GHG inventory

Why – why firms bother with GHG inventory?

Internal

External

What – what is counted?

A defined geographic boundary

Gases, sectors

Reporting year

How – how is it quantified?

Activity Data * Emission Factors

Activity data:

Emission factors:

So What – how to interpret it?

Comparability

Vs

Reliability

NEXT: Product level carbon accounting

Next Time – Product Level Carbon Accounting

- **Essential reading:**
  - GHG Protocol (2011). Product Life Cycle Accounting and Reporting Standard.
    - For Lecture 9: read Chapter 6, pp. 28–30, and examine Figures 7.1 (p. 34) and 7.2 (p. 36).Focus on the functional unit, life-cycle stages and system boundary.
  - Lu S., Serafeim G. and Tofffel M.(2022). Driving Decarbonization at BMW. HBS 9-123-008. https://hbsp.harvard.edu/product/
    - For Lecture 10: read pp. 10–12 and examine Exhibits 14a (p. 29) and 15 (p. 30).
      Focus on materials, batteries, vehicle efficiency and electricity assumptions.
- **Recommended reading**
  - GHG Protocol (2011). Product Life Cycle Accounting and Reporting Standard.
    - Chapter 8, pp. 47–48 and Table 8.1 (p. 55): data types and quality.
      Chapter 9, pp. 61–63: why allocation is needed and the main approaches.
- **Further reading:**
  - ISO 14067:2018. Requirements and guidelines for product carbon footprint quantification.